



Loan Analytics in DROP

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The Distribution of Loan Portfolio Value

Introduction and Overview

1. Portfolio Debt Securities Loss Capital: The amount of capital necessary to support debt securities depends on the probability distribution of the portfolio loss.
2. Portfolio Subject to Default Loss: Consider a portfolio of loans, each of which is subject to default resulting in a loss to the lender.
3. Financing Split of the Portfolio: Suppose that the portfolio is financed partly by equity capital and partly by borrowed funds.
4. Credit Quality of the Portfolio Notes: The credit quality of the lender's notes will depend on the probability that the loss on the portfolio exceeds the equity capital.
5. Maintenance of the Notes' Rating: To achieve a certain credit rating of its notes – say Aa on a rating agency scale – the lender needs to keep the probability of default on the notes at the level corresponding to that rating – about 0.001 for the Aa quality.
6. Estimating the Corresponding Equity Capital: This means that the equity capital allocated to the portfolio must be equal to the percentile of the distribution of the portfolio loss that corresponds to the desired probability.
7. Application of Portfolio Loss Distribution: In addition to determining the capital necessary to support a loan portfolio, the probability distribution of portfolio losses has a number of other applications. It can be used in regulatory reporting, measuring portfolio risk, calculating the VaR, portfolio optimization, and structuring and pricing of debt portfolio derivatives such as Collateralized Debt Obligations (CDO).
8. Primary Focus of this Chapter: Following Vasicek (2002), this chapter derives the portfolio loss distribution under certain assumptions.



9. Portfolio Size Loss Distribution Dependence: It is shown here that the distribution converges with increasing portfolio size to a limiting type, whose analytical form is given here.
10. Literature Related to Loss Distribution: The results of the first two sections of this chapter are contained in the technical notes provided by Vasicek (1987, 1991). For a review of literature on the subject, see, for instance, Pykhtin and Dev (2002).

The Limiting Distribution of Portfolio Losses

1. Loan Borrower Asset Value Model: Assume that a loan defaults if the value of the borrower's assets at the loan maturity T falls below the contractual value B of its obligations.
2. Loan Borrower Asset Value Dynamics: Let A_i be the value of the i^{th} borrower's assets, described by the process

$$\Delta A_i = \mu_i A_i \Delta t + \sigma_i A_i \Delta x_i$$

3. Evolution of the Asset Value: The asset value at T can be represented as

$$\log A_i(T) = \log A_i(0) + \mu_i T - \frac{1}{2} \sigma_i^2 T + \sigma_i \sqrt{T} X_i$$

where X_i is a standard normal variable.

4. Default Probability for Loan i : The probability of default of the i^{th} loan is then



$$p_i = \mathbb{P}[A_i(T) < B_i] = \mathbb{P}[X_i < c_i] = \Phi(c_i)$$

where

$$c_i = \frac{\log B_i - \log A_i(0) - \mu_i T + \frac{1}{2} \sigma_i^2 T}{\sigma_i \sqrt{T}}$$

and Φ is the cumulative normal distribution function.

5. Homogenous Portfolio Loan Component Specification: Consider a portfolio consisting of n loans in equal dollar amounts. Let the probability of default of any one loan be p , and assume that the asset values of the borrowing companies are correlated with a coefficient ρ for any two companies. It is further assumed that all loans have the same term T .
6. Single and Portfolio Loan Loss: Let L_i be the gross loss before recoveries on the i^{th} loan so that

$$L_i = 1$$

if the i^{th} borrower defaults and

$$L_i = 0$$

otherwise. Let L be the portfolio percentage gross loss



$$L = \frac{1}{n} \sum_{i=1}^n L_i$$

7. Independence of Component Loan Losses: If the events of the defaults of the loans in the portfolio were independent of each other, the portfolio loss distribution would converge, by the central limit theorem, to a normal distribution as the portfolio size increases.
8. Convergence of the Portfolio Loss: Because the defaults are not independent, however, the conditions of the central limit theorem are not satisfied, and L is not asymptotically normal. It turns out, however, that the distribution of the portfolio loss does converge to a limiting term, which shall now be derived.
9. Systemic and Idiosyncratic Factor Decomposition: The variables X_i in

$$\log A_i(T) = \log A_i(0) + \mu_i T - \frac{1}{2} \sigma_i^2 T + \sigma_i \sqrt{T} X_i$$

are jointly standard normal with equal pair-wise correlation ρ , and can therefore be represented as

$$X_i = Y\sqrt{\rho} + Z_i\sqrt{1-\rho}$$

where $Y, Z_1, \dots, Z_n$ are mutually independent standard normal variables. This is not an assumption, but a property of the equi-correlated normal distribution.

10. Y as Systemic Common Factor: The variable Y can be interpreted as a portfolio common factor, such as an economic index, over the interval $(0, T)$.



11. Z_i as Idiosyncratic Individual Factor: The term $Y\sqrt{\rho}$ is the company's exposure to the common factor and the term $Z_i\sqrt{1-\rho}$ represents the company specific risk.
12. Portfolio Loss Conditional on Y : The probability of the portfolio loss as an expectation over the common factor Y will now be evaluated given the conditional probability Y .
13. Interpretation of the Conditional Portfolio Loss: This can be interpreted as assuming various scenarios for the economy, determining the probability of a given portfolio loss under each scenario, and weighting each scenario by its likelihood.
14. Single Loan Conditional Default Probability: When the common factor is fixed, the conditional probability of loss on any one loan is

$$p(Y) = \mathbb{P}[L_i = 1|Y] = \Phi\left(\frac{\Phi^{-1}(p) - Y\sqrt{\rho}}{\sqrt{1-\rho}}\right)$$

15. Single Loan Unconditional Default Probability: The quantity $p(Y)$ provides the loan default probability under the given scenario. The unconditional default probability p is just the average of the conditional probabilities over all the scenarios.
16. Conditional on Y Distribution of L_i : Conditional on the value of Y , the variables X_i are independent and equally distributed with a finite variance.
17. Applying CLT to Large Portfolio: The portfolio loss conditional on Y converges, by the law of large numbers, to its expectation $p(Y)$ as

$$n \rightarrow \infty$$

18. Unconditional Large Portfolio Loss Distribution: Then



$$\mathbb{P}[L \leq x] = \mathbb{P}[p(Y) \leq x] = \mathbb{P}[Y \geq p^{-1}(x)] = \Phi(-p^{-1}(x))$$

and, on substitution, the cumulative distribution function of loan losses on a very large portfolio is in the limit

$$\mathbb{P}[L \leq x] = \Phi\left(\frac{\sqrt{1-\rho}\Phi^{-1}(x) - \Phi^{-1}(p)}{\sqrt{\rho}}\right)$$

This result is given by Vasicek (1991).

19. Portfolio Weights with Unequal Components: The convergence of the portfolio loss distribution to the limiting form above actually holds even for portfolios with unequal weights.
20. Applying CLT to this Portfolio: Let the portfolio weights be $w_1, \dots, w_n$ with $\sum_{i=1}^n w_i$ the portfolio loss

$$L = \sum_{i=1}^n w_i L_i$$

conditional on Y conditional on Y converges to its expectation $p(Y)$ whenever – and this is a necessary and sufficient condition –

$$\sum_{i=1}^n w_i^2 \rightarrow 0$$



21. Interpretation of the Weight Condition: In other words, if the portfolio contains a sufficiently large number of loans without it being dominated by a few loans much larger than the rest, the limiting distribution provides a good approximation for the portfolio loss.

Properties of the Loss Distribution

1. Support of the Portfolio Loss PDF: The portfolio loss distribution given by the cumulative distribution function

$$F(x; p, \rho) = \Phi \left(\frac{\sqrt{1 - \rho} \Phi^{-1}(x) - \Phi^{-1}(p)}{\sqrt{\rho}} \right)$$

is a continuous distribution concentrated on the interval

$$0 \leq x \leq 1$$

2. Two Parameter Family of Distribution: It forms a two-parameter family with the parameters

$$p > 0$$



and

$$\rho < 1$$

3. Limiting Behavior for ρ Values: When

$$\rho \rightarrow 0$$

it converges to a one-point distribution concentrated at

$$L = p$$

When

$$\rho \rightarrow 1$$

it converges to a zero-one distribution with probabilities p and $1 - p$, respectively.

4. Limiting Behavior for p Values: When

$$p \rightarrow 1$$



the distribution becomes concentrated at

$$L = 0$$

and

$$L = 1$$

respectively.

5. Symmetry Property of the Distribution: The distribution possesses a symmetry property

$$F(x; p, \rho) = 1 - F(1 - x; 1 - p, \rho)$$

6. Density of the Loss Distribution: The loss distribution has the density

$$f(x; p, \rho) = \sqrt{\frac{1 - \rho}{\rho}} e^{-\frac{1}{2\rho} \{ [\sqrt{1 - \rho} \Phi^{-1}(x) - \Phi^{-1}(p)]^2 - \frac{1}{2} [\Phi^{-1}(x)]^2 \}}$$

which is unimodal at



$$L_{MODE} = \Phi \left(\frac{\sqrt{1-\rho}}{\sqrt{1-2\rho}} \Phi^{-1}(p) \right)$$

when

$$\rho < \frac{1}{2}$$

monotone when

$$\rho = \frac{1}{2}$$

and U-shaped when

$$\rho > \frac{1}{2}$$

7. Mean/Variance of the Loss Distribution: The mean of the distribution is

$$\mathbb{E}[L] = p$$

and the variance is



$$s^2 = \mathbb{V}[L] = \Phi_2(\Phi^{-1}(p), \Phi^{-1}(p), \rho)$$

where Φ_2 is the bivariate cumulative normal distribution function.

8. Usage in Cumulative p-Value Tables: The inverse of this distribution, that is, the percentile value of L , is given by

$$L_\alpha = 1 - F(\alpha; 1 - p, \rho)$$

9. Skew and Kurtosis of the Distribution: The portfolio loss distribution is highly skewed and leptokurtic. The table below lists the α -percentile L_α expressed as a number of the standard deviations from the mean, for several values of the parameters. The α -percentiles of the standard normal distribution are shown for comparison.
10. Values of α -percentiles for the Portfolio Loss Distribution:

p	ρ	$\alpha = 0.9000$	$\alpha = 0.9900$	$\alpha = 0.9990$	$\alpha = 0.9999$
0.010	0.1	1.19	3.80	7.00	10.70
0.010	0.4	0.55	4.50	11.00	18.20
0.001	0.1	0.98	4.10	8.80	15.40
0.001	0.4	0.12	3.20	13.20	31.80
NORMAL		1.28	2.30	3.10	3.70



11. Non-normality of the Distribution: These values manifest the extreme non-normality of the loss distribution.
12. Sample Large Portfolio Loss Distribution: Suppose a lender holds a large portfolio of loans to firms whose pair-wise asset correlation is

$$\rho = 0.4$$

and whose probability of default is

$$p = 0.01$$

The expected portfolio loss is

$$\mathbb{E}[L] = 0.01$$

and the standard deviation is

$$s = 0.0277$$

If the lender wishes to hold the probability of default on his notes at



$$1 - \alpha = 0.001$$

he will need enough capital to cover 11.0 times the portfolio standard deviation. If the loss were normal, 3.1 times the standard deviation would suffice.

The Risk-Neutral Distribution

1. Actual World Loss Distribution Measure: The portfolio loss distribution given by

$$\mathbb{P}[L \leq x] = \Phi \left(\frac{\sqrt{1 - \rho} \Phi^{-1}(x) - \Phi^{-1}(p)}{\sqrt{\rho}} \right)$$

is the actual probability distribution.

2. Probability of the Specified Loss Realization: This is the probability distribution from which to calculate the probability of loss of a certain magnitude for the purposes of determining the necessary capital or for calculating VaR.
3. Use in Structuring of CDO's: This is the distribution to be used in structuring collateralized debt obligations; that is calculating the probabilities of loss and the expected loss for a given tranche.
4. Risk-Neutral World Asset Dynamics: For the purposes of pricing the tranches, however, it is necessary to use the risk-neutral probability distribution. The risk-neutral distribution is calculated in the same way as above, except that the default probabilities are calculated under the risk-neutral measure



$$p^* = \mathbb{P}^*[A(T) < B] = \Phi\left(\frac{\log B - \log A - rT + \frac{1}{2}\sigma T^2}{\sigma\sqrt{T}}\right)$$

where r is the risk-free rate.

5. Risk-Neutral vs. Actual World: The risk-neutral probability is related to the actual probability of default by the equation

$$p^* = \Phi(\Phi^{-1}(p) + \lambda\rho_M\sqrt{T})$$

where ρ_M is the correlation of the firm asset value with the market and

$$\lambda = \frac{\mu_M - r}{\sigma_M}$$

is the market price of risk.

6. Risk-Neutral Loss Distribution Measure: The risk-neutral portfolio loss distribution is then given by

$$\mathbb{P}^*[L \leq x] = \Phi\left(\frac{\sqrt{1-\rho}\Phi^{-1}(x) - \Phi^{-1}(p^*)}{\sqrt{\rho}}\right)$$



7. Risk-Neutral Derivative Pricing: Thus, a derivative security – such as CDO tranche written against a portfolio – That pays at time T and amount $C(L)$ contingent on the portfolio loss is valued at

$$V = e^{-rT} \mathbb{E}^*[C(L)]$$

where the expectation is taken with respect to the distribution

$$\mathbb{P}^*[L \leq x] = \Phi\left(\frac{\sqrt{1-\rho}\Phi^{-1}(x) - \Phi^{-1}(p^*)}{\sqrt{\rho}}\right)$$

8. Tranche Pricing under this Measure: For instance, a default protection for losses in excess of L_0 is priced at

$$V = e^{-rT} \mathbb{E}^*[L - L_0] = e^{-rT} \{p^* - \Phi_2(\Phi^{-1}(p^*), \Phi^{-1}(L_0), \sqrt{1-\rho})\}$$

The Portfolio Market Value

1. Forward Value for Expected Loss: So far, the treatment has been about the loss due to loan defaults. Now, suppose that the maturity date T_M is past the date T_H for which the portfolio is examined, i.e., the horizon date.



2. Impact of the Borrower Credit Quality: If the credit quality of the borrower deteriorates, the value of the loan will decline, resulting in a loss – this is often referred to as the loss due to *credit migration*.
3. MTM Change Impact on Portfolio Loss: This section investigates the distribution of the loss resulting from the changes in the MTM portfolio value.
4. Risk-Neutral Value of the Loan: The value of the loan at time 0 is the expected value of the loan payments under the risk-neutral measure

$$D = e^{-rT_M}(1 - Gp^*)$$

where G is the log given default, and p^* is the risk-neutral probability of default.

5. Value of the Loan at T_H : At time T_H the value of the loan is

$$D(T_H) = e^{-rT_H} \left\{ 1 - G\Phi \left(\frac{\log B - \log A(T_H) - r[T_M - T_H] + \frac{1}{2}\sigma[T_M - T_H]^2}{\sigma\sqrt{T_M - T_H}} \right) \right\}$$

6. Value of the Loss at T_H : Defining the loan loss L_i at time T_H as the difference between the risk-less value and the market value of the loan at T_M one gets

$$L_i = e^{-r[T_M - T_H]} - D(T_H)$$

7. Alternative Definition for Loan Loss: This definition for loss is chosen purely for convenience. If the loss is defined in a different way – for instance, as a difference between



the accrued value and the market value – it will only result in a shift of portfolio loss distribution by a location parameter.

8. Explicit Expression for Loan Loss: The loss on the i^{th} loan can be written as

$$L_i = a\Phi\left(b\sqrt{\frac{T_M}{T_M - T_H}} - X_i\sqrt{\frac{T_M}{T_M - T_H}}\right)$$

where

$$a = Ge^{-r[T_M - T_H]}$$

$$b = \Phi^{-1}(p) + \lambda\rho_M \frac{T_M - T_H}{T_M}$$

and the standard normal variables X_i defined over the horizon T_H by

$$\log A_i(T_H) = \log A_i(0) + \mu_i T_H - \frac{1}{2}\sigma_i^2 T_H + \sigma_i\sqrt{T_H}X_i$$

are subject to

$$X_i = Y\sqrt{\rho} + Z_i\sqrt{1 - \rho}$$



9. Explicit Expression for Condition Loss: The conditional mean of L_i given Y can be calculated as

$$\mu(Y) = \mathbb{E}[L_i|Y] = a\Phi\left(b\sqrt{\frac{T_M}{T_M - \rho T_H}} - Y\sqrt{\frac{\rho T_H}{T_M - \rho T_H}}\right)$$

10. Convergence of Portfolio Expected Loss: Let L be the market value loss at time T_H of a loan portfolio with weights w_i . The losses conditional on factor Y are independent, and therefore the portfolio loss L conditional on Y converges to its mean value

$$\mathbb{E}[L|Y] = \mu(Y)$$

as

$$\sum_{i=1}^n w_i^2 \rightarrow \infty$$

11. Limiting Distribution of Portfolio Loss: The limiting distribution of loss L is then

$$\mathbb{P}[L \leq x] = F\left(\frac{x}{a}; \Phi(b), \rho \frac{T_H}{T_M}\right)$$



12. Form of the Limiting Distribution: It can thus be seen that the limiting loss of the portfolio distribution is of the same type as

$$F(x; p, \rho) = \Phi \left(\frac{\sqrt{1-\rho}\Phi^{-1}(x) - \Phi^{-1}(p)}{\sqrt{\rho}} \right)$$

whether the loss is defined as a decline in the market value or the realized loss at maturity. In fact, the results of the section on the distribution of loss due to default are just a special case of this section for

$$T_M = T_H$$

13. Risk-Neutral Measure for MTM: The risk-neutral distribution for the loss due to market value change is given by

$$\mathbb{P}^*[L \leq x] = F \left(\frac{x}{a}; p^*, \rho \frac{T_H}{T_M} \right)$$

Adjustment for Granularity



1. Zero Variance Criterion for CLT: The limiting portfolio loss distribution of the loss L is then

$$\mathbb{P}[L \leq x] = \mathbb{P}[\mu(Y) \leq x] = F\left(\frac{x}{a}; \Phi(b), \rho \frac{T_H}{T_M}\right)$$

relies on the convergence of the portfolio loss L given Y to its mean value $\mu(Y)$, which means that the conditional variance

$$\mathbb{V}[L|Y] \rightarrow 0$$

2. Adjustment for Non-zero Variance: When the portfolio is not sufficiently large for the law of large numbers to take hold, one needs to take into account the non-zero value of $\mathbb{V}[L|Y]$.
3. Sum of Quadratic Weights: Consider a portfolio of uniform credits with weights $w_1, \dots, w_n$ and set

$$\delta = \sum_{i=1}^n w_i^2$$

4. Conditional Variance of the Portfolio Loss: The conditional variance of the portfolio loss L given Y is

$$\mathbb{V}[L|Y] = \delta a^2 \left\{ \Phi_2 \left(\Phi^{-1}(U), \Phi^{-1}(U), \sqrt{\frac{(1-\rho)T_H}{T_M - \rho T_H}} \right) - \Phi^2(U) \right\}$$



where

$$U = b \sqrt{\frac{T_M}{T_M - \rho T_H}} - Y \sqrt{\frac{\rho T_H}{T_M - \rho T_H}}$$

5. Unconditional Portfolio Mean and Variance: The unconditional mean and variance of the portfolio loss are

$$\mathbb{E}[L] = a\Phi(b)$$

and

$$\begin{aligned} \mathbb{V}[L] &= \mathbb{V}[\mathbb{E}[L|Y]] + \mathbb{E}[\mathbb{V}[L|Y]] \\ &= \delta a^2 \Phi_2\left(b, b, \frac{T_H}{T_M}\right) + (1 - \delta) a^2 \Phi_2\left(b, b, \rho \frac{T_H}{T_M}\right) - a^2 \Phi^2(b) \end{aligned}$$

6. Approximation of the Unconditional Variance: Taking the first two terms in the tetrachoric expansion of the bivariate normal distribution function

$$\Phi_2(x, x, \rho) = \Phi^2(x) + \rho \phi^2(x)$$



where ϕ is the normal density function, one has approximately

$$\mathbb{V}[L] = \delta a^2 \phi^2(b) \frac{T_H}{T_M} + a^2 \Phi_2 \left(b, b, \{\rho + \delta[1 - \rho]\} \frac{T_H}{T_M} \right) - a^2 \Phi^2(b)$$

7. Moment Matching Loss Distribution Fit: Approximating the loan loss distribution by

$$F(x; p, \rho) = \Phi \left(\frac{\sqrt{1 - \rho} \Phi^{-1}(x) - \Phi^{-1}(p)}{\sqrt{\rho}} \right)$$

with the same mean and variance, one gets

$$\mathbb{P}[L \leq x] = F \left(\frac{x}{a}; \Phi(b), \{\rho + \delta[1 - \rho]\} \frac{T_H}{T_M} \right)$$

8. Asymptotic Behavior for n, δ : This expression is in fact exact for both extremes

$$n \rightarrow \infty$$

$$\delta = 0$$

and



$$n = 1$$

$$\delta = 1$$

9. Adjustment for the Granularity of Portfolio: The loss distribution

$$\mathbb{P}[L \leq x] = F\left(\frac{x}{a}; \Phi(b), \{\rho + \delta[1 - \rho]\} \frac{T_H}{T_M}\right)$$

provides an adjustment for the *granularity* of the portfolio. In particular, the finite portfolio adjustment to the distribution of the gross loss at the maturity date is obtained by setting

$$T_H = T_M$$

and

$$a = 1$$

to yield



$$\mathbb{P}[L \leq x] = F(x; \Phi(b), \rho + \delta[1 - \rho])$$

Summary

1. Convergence of Portfolio Loss Distribution: This chapter has shown that the distribution of loan portfolio loss converges, with increasing size, to the limiting type given by

$$F(x; p, \rho) = \Phi\left(\frac{\sqrt{1 - \rho}\Phi^{-1}(x) - \Phi^{-1}(p)}{\sqrt{\rho}}\right)$$

This means that the distribution can be used to represent loan loss behavior of large portfolios.

2. Spot/Forward Pricing of Loss: The loan loss can be a realized loss on the loans maturing prior to the horizon date, or a market value deficiency on the loans whose term is longer than the horizon period.
3. Derivation of the Limiting Loss Distribution: The limiting probability distribution of the portfolio losses has been derived under the assumption that all loans in the portfolio have the same maturity, the same probability of default, and the same pair-wise correlation of borrower assets.
4. Applicability to more General Portfolios: Surprisingly, however, numerical simulations show that the family

$$F(x; p, \rho) = \Phi\left(\frac{\sqrt{1 - \rho}\Phi^{-1}(x) - \Phi^{-1}(p)}{\sqrt{\rho}}\right)$$



appears to provide a reasonably good fit to the tail of the loss distribution for more general portfolios.

5. Comparison with Monte Carlo Simulations: To illustrate this point, Vasicek (2002) provide the results of Monte-Carlo simulations on an actual bank portfolio.
6. Loan Portfolio Characteristics - Component Weight: The portfolio consisted of 479 loans in amounts ranging from 0.0002% to 8.7% with

$$\delta = 0.039$$

7. Loan Portfolio Characteristics - Maturity/PD: The maturities ranged from 6 months to 6 years and the default probabilities from 0.0002 to 0.064.
8. LGD + Asset Returns Common Factors: The loss give default averaged 0.54. The asset returns were generated with 14 common factors.
9. Simulated vs. Analytical CDF Comparison: Vasicek (2002) compares in detail the simulated cumulative distribution function of the loss in one year against the fitted limiting distribution function.

References

- Pykhtin, M. and A. Dev (2002): Credit Risk in Asset Securitizations: An Analytical Model *Risk* **15 (5)** 16-20
- Vasicek (1987): *Probability of Loss on Loan Portfolio* **KMV Corporation**
- Vasicek (1991): *Limiting Loan Loss Probability Distribution* **KMV Corporation**



- Vasicek (2002): [The Distribution of Loan Portfolio Value](#)



Vasicek Model Default Risk Simulation

Theoretical Background

1. Portfolio Loss Distribution Model Used: One-factor Gaussian Copula Credit Loss Model.
2. Incorporation of Asset Default Correlation: Default is driven by a common market factor Y - the systemic shock – and an idiosyncratic Gaussian Process Z_i :

$$X_i = Y\sqrt{\rho} + Z_i\sqrt{1-\rho}$$

3. Vasicek Expression for Loss Distribution: The famous Vasicek closed-form formula for calculating the loss distribution under the Gaussian Copula framework:

$$VaR_\alpha = \Phi\left(\frac{\Phi^{-1}(p) + \sqrt{\rho}\Phi^{-1}(\alpha)}{\sqrt{1-\rho}}\right)$$

4. Assumptions of the Vasicek Model:
 - All exposures are of the same amount – homogenous portfolio
 - All exposures have the same probability of default



- All exposures have a common set correlation

Model Implementation

1. Correlation to the Common Factor:

$$Factor_1 = \sqrt{\rho}$$

2. Correlation to Idiosyncratic Factor:

$$Factor_2 = \sqrt{1 - \rho}$$

3. Default Threshold:

$$d_t = \Phi^{-1}(1 - p)$$

4. Systemic Gaussian Process:

$$Y = \Phi^{-1}(rand(\cdot))$$



5. Idiosyncratic Gaussian Process:

$$Z_i = \Phi^{-1}(\text{rand}(\cdot))$$

6. Asset Value Default Incidence Simulation: If

$$Factor_1 \cdot Y + Factor_2 \cdot Z_i > d_t$$

the exposure is treated as having defaulted.

7. Default Simulation across Counter-Parties: The systemic and the idiosyncratic variables for many paths are simulated across counter-parties, and for each, the total loss is summed in that exposure.
8. Risk Capital as Target Loss Percentile: The risk capital is the picked for the target percentile from these sums of total losses across all the simulation paths.

Sample Simulation Results

1. Uniform Exposure across all Ratings:

Uniform 1MM Exposure Size	Confidence	90%
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Row Labels	No Hedge Cumulative Exposure	2Y PD	Correlation	Tail Loss
1	265,000,000	0.001050	0.238213433	661,866
2	264,000,000	0.004926	0.231836920	3,305,600
3	311,000,000	0.003169	0.231836920	2,477,183
4	276,000,000	0.012852	0.215292799	8,937,681
5	311,000,000	0.065197	0.160048020	42,807,422
6	313,000,000	0.189831	0.121497866	100,943,573
7	260,000,000	0.481051	0.120000001	172,562,865
Grand Total	2,000,000,000			331,696,209

Average of Simulations	332,687,000
Simulation Difference Standard Deviation	6,637,385
Simulation Difference Standard Deviation %	2.00%
# > 0 Difference %	56
# < 0 Difference %	44

- Upper table is the closed for result, bottom table is the simulated result.
- As expected, the two results are close.



2. Two Outsized Exposures in AA:

Uniform 1MM Exposure Size			Confidence	90%
Row Labels	No Hedge Cumulative Exposure	2Y PD	Correlation	Tail Loss
1	265,000,000	0.001050	0.238213433	661,866
2	18,262,000,000	0.004926	0.231836920	228,662.386
3	311,000,000	0.003169	0.231836920	2,477,183
4	276,000,000	0.012852	0.215292799	8,937,681
5	311,000,000	0.065197	0.160048020	42,807,422
6	313,000,000	0.189831	0.121497866	100,943,573
7	260,000,000	0.481051	0.120000001	172,562,865
Grand Total	19,998,000,000			557,052,995

Average of Simulations	334,552,000
Simulation Difference Standard Deviation	6,696,572
Simulation Difference Standard Deviation %	1.26%
# > 0 Difference %	0
# < 0 Difference %	100



- Upper table is the closed for result, bottom table is the simulated result.
- Closed form doesn't work well in such a case.

3. Two Bigger Exposures in B:

Uniform 1MM Exposure Size			Confidence	90%
Row Labels	No Hedge Cumulative Exposure	2Y PD	Correlation	Tail Loss
1	265,000,000	0.001050	0.238213433	661,866
2	264,000,000	0.004926	0.231836920	3,305,600
3	311,000,000	0.003169	0.231836920	2,477,183
4	276,000,000	0.012852	0.215292799	8,937,681
5	311,000,000	0.065197	0.160048020	42,807,422
6	701,000,000	0.189831	0.121497866	226,074,903
7	260,000,000	0.481051	0.120000001	172,562,865
Grand Total	2,000,000,000			456,827,539

Average of Simulations	514,818,000
Simulation Difference Standard Deviation	12,685,022
Simulation Difference Standard Deviation %	2.78%
# > 0 Difference %	100



# < 0 Difference %	0
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- Upper table is the closed for result, bottom table is the simulated result.
- Simulated rating captures more idiosyncratic risk than the closed-form, indicating the importance of PD (rating) of the exposure.

Implication of the Simulation Tests

1. Addressing Vasicek Drawbacks using Simulation:

- Vasicek expression cannot be used blindly to get the systemic default risk
- Alternative ways of estimating systemic default risk can be done by breaking down portfolios into infinite number of smaller exposures and run large simulation
- Drawback is that the implementation will be more complex and the run more time consuming

2. HHI as Portfolio Homogeneity Metric:

- One indicator can help test the homogeneity of the portfolio
- Introducing the Herfindahl-Hirschmann Index (HHI)

$$HHI = \sum_{i=1}^n \left(\frac{EAD_i}{\sum_{j=1}^n EAD_j} \right)^2 \rightarrow 0$$

as



$$n \rightarrow \infty$$

- For an infinitely large homogenous portfolio, the HHI should approach 0
- Our exposure without hedge HHI is 0.9% with hedge is 0.4%

Limitations of the Model

1. Model Parameter Calibration Limitation Description: in regards to calibration of model parameters, the probabilities of default (PD's) and the default correlations are calibrated using the Vasicek (2002) Credit Model whereas the Loss Given Default (LGD's) are calibrated using historical observations. The calibration of LGD's and PD's is based on limited historical data and the assumption of the homogenous base portfolio. The calibrated model inputs are difficult to observe and may change with time. The model parameters may also differ between alternative energy sources or sampling methodologies.
2. Model Parameter Calibration Severity Assessment: The impact of this limitation on the model parameter is limited. The model will be replaced in the near future by a six-factor model similar to the one commonly in use in CCAR IDL.
3. Model Parameter Calibration Remediation Action: The calibration of the model inputs, i.e., PD's and LGD's, should be benchmarked with the historical data available from third party vendor's such as S&P's and Moody's on an ongoing annual basis as part of Ongoing Performance Analysis (OPA) and the calibration process for the model parameters should be reviewed.



4. Ratings Based Calibration Limitation Description: The PD's constitute a key model input, provided by the BHC credit model. Default probabilities are often calculated by risk rating so that all names in the corresponding ratings category take the expected default probability. As a result, the default probabilities may not be distinguished between industries and sectors. In addition, the calculation of the default risk for the CVA is based on a single factor Gaussian copula for the estimation of the correlation among the counter-parties. While this may be sufficient for the Risk Capital (RC) estimation as an industry standard, it is a simplistic approach. As a result the model may not capture the diversity of different markets and industry sectors. Also, it may not reflect the accurate correlation between the counter-parties.
5. Ratings Based Calibration Severity Assessment: This limitation is related to model framework and assessment. The impact on the model output is potentially moderate. A multi-factor model, similar to the one often used in CCAR IDL, is a candidate for replacing this.
6. Ratings Based Assessment Remediation Action: The multi-factor model, which is capable of diversifying the correlation between different industries, sectors, and regions, is a viable candidate for replacement. More granular and industry representative default probabilities may be considered in the model development.
7. Input Sensitivity Tests Limitation Description: To further assess the efficacy and the impact of the model, the following sensitivity analysis must be performed, preferably on a recently available data:
 - Evaluate the changes in the risk capital with respect to the PD's.
 - Evaluate the changes in the risk capital with respect to the changes in the default correlation input parameters.
8. Input Sensitivity Tests Severity Assessment: This limitation is to recognize more tests, and does not indicate model weakness. A multi-factor mode, similar to the one used in CCAR IDL, is a candidate for replacing this. Additional tests will need to be performed for this new model.
9. Input Sensitivity Tests Remediation Action: The sensitivity tests should be performed against the mode input parameters – e.g., PD's and different correlations – on a recently available data, and the performance should be assessed.



References

- Vasicek (2002): [The Distribution of Loan Portfolio Value](#)



Market Place Lending Credit Model Methodology

Overview of Credit Model Methodology

1. Risk Estimation in MPL Platforms: DROP (2015) has introduced version 2.0 of the Credit Model for estimating prepayment and default risk on a pool of US unsecured consumer loans issued by a select set of marketplace lending (“MPL”) platforms.
2. Valuation, Prepay, and Default Drivers: This extends the version 1.0 Credit Model published previously in August 2015. In both cases the goal is to enable the market participants understand the drivers behind valuation, prepayment, and default risk across their holdings in a transparent and robust manner.
3. Usage of Markov Logit Models: A main objective is to apply the rigor and forward-looking tools of mortgage analytics, namely the Multinomial Logit and the Markov Chain approach to modeling the MPL collateral.
4. Simulation of Prepay and Default: The second objective is to have the ability apply simulations to prepayment and default rates, and ultimately the cash flows.
5. Usage of Statistical Learning Techniques: The final objective is to use machine learning techniques such as regularization and cross-validation to improve the robustness of the modeling.

Credit Methodology – Purpose and Introduction

1. MPL Growth/Institutional Investors Participation: The dramatic growth in Marketplace Lending (MPL) has been paired with the transition of the consumer and the SME credit to the



capital markets. Institutional investors are increasingly funding credit risk in whole loan, structured credit, or warehouse formats.

2. MPL Risk and Analytics Tools: A new set of analytics and risk pricing tools are necessary to enable institutional capital efficiently access, price, and exchange risk. A core component of risk pricing is independent 3rd party credit models that forecast cash flows on pools of loans.
3. Improvement in Liquidity and Transparency: By releasing a new credit model, the aim is to improve transparency and standards in MPL by providing risk management tools promoting independent pricing and whole loan and ABS market liquidity.
4. Valuation and Risk MPL Common Language: Independent credit models increase activity by allowing market participants to speak a common language in reference to valuation and risk.
5. Independent 3rd Party Credit Models: Several sources have described on various occasions the robust scaling and growth in the US and global marketplace lending sector. Large institutional investors require a 3rd party credit model to understand their credit risk exposures, and to participate in size.
6. Short Duration, High Yield Risk: Further, credit spreads have generally tightened since the 2008 credit crisis in a backdrop of quantitative easing, healthy global economy, and stringent regulation. As a result, investors see MPL offering an attractive short-duration, high-yield credit risk compared to the alternatives.
7. Engagement in the MPL Space: Investors have been engaged in the market through a variety of activities including, but not limited to:
 - a. Investing in marketplace lending platform equity
 - b. Facilitating funding for the platforms via a provision of credit facilities
 - c. Investing in MPL securitizations
 - d. Directly lending to borrowers.
8. Proliferation among MPL Asset Classes: In addition, there has been a proliferation of marketplace lenders across a multitude of asset classes including consumer, purchase finance, education finance, real estate, merchant cash advance, and small businesses.
9. Asset Pricing and Regulation Obligation: As a consequence, institutional investors, diversified financial services firms, and funding providers require an independent 3rd party to help them price their holdings or satisfy other regulatory obligations.



10. MPL Valuation Standards Methodology Enhancement: Finally, it is imperative that 3rd parties commit to improving and enhancing their standards and methodologies to serve what is a rapidly growing and evolving market. The 2.0 Credit Model is an attempt at enhancements over several key areas for projections of cash flows on historical data above 1.0.
11. Additional Loan Specific Risk Factors: 2.0 addresses the need to incorporate additional loan specific measures of risk. Version 1.0 segments the loan by 6 factors:
 - a. The originator
 - b. Loan credit quality grade as provided by the originator
 - c. Origination vintage
 - d. Loan term
 - e. Loan status
 - f. Loan age
12. Incorporating Macro-economic Driving Factors: 2.0 provides a structure for incorporating macro-economic factors that drive the estimates of default and prepayment.
13. Reducing Dependence on Recent Issuance: Reducing the reliance on the most recently issued set of loans to drive expectation of prepay and default is another objective (1.0 applied the default and prepay experience of the most recently issued cohort corresponding to the risk factors above).
14. Usage of Statistical Learning Techniques: Version 2.0 applies advanced statistical and machine learning techniques to develop a predictive model for prepay and default. Thus, Credit Model Version 2.0 aims to address and overcome the shortcomings from the Model 1.0 above, and bring rigorous techniques to an expanding asset class with a growing investor base.

Scope of the 2.0 Model



1. Loans Originated by Lending Club: For the purposes of demonstration, PeerIQ (2015) illustrate the construction and performance of Model Version 2.0 on public data from loans originated by Lending Club (“LC”) with reporting months from 1 January 2010 to 1 July 2015.
2. The Data Model: PeerIQ’s data model is proprietary and unified in its methodology for cleaning, enriching, and housing data across all MPL originators, and gets expanded as additional asset classes and originators are on-boarded.
3. Similarity with Lending Club Model: Although DROP does tailor the Model to specific data classes and originators, the methodology and the model structure is substantially similar to the Lending Club model.

Data Model Construction Rules

1. Amendments to Originator Generated Payments: DROP has made specific amendments (or transformations) on the raw originator-generated payments and balances for loans.
2. Consistency and Accuracy across Cohorts: These changes have been made in a rule-based fashion based on conversations with marketplace lenders to ensure the calculation of the cohort payments and balances in a consistent and accurate manner.
3. Reconciliation between Borrowers and Originators: Many of the above rules help reconcile between a borrower snapshot file (the borrower file) and a cumulative payments file (the payments file) published by the originators.

Loan Data Quality Rules



1. Inaccurate Originator Loan Level Record: Some originators publish inaccurate records for loans that have previously charged or paid off. These records are excluded from the cleaned data set and calculations.
2. Identification and Removal of Duplicates: If the loan has more than one record with the same originator loan ID, loan month, month on book, and outstanding principal BOP balance, it is assumed that the subsequent records are duplicates and that they must be removed.
3. Combine Payments for a Given Month: If the loan has multiple payments for a given month, then these must be combined to form a single payment. The formula applied is: combine all rows where count of loan month, originator loan ID > 1.
4. Entry for Maximum Loan Month: Each loan that has a non-zero EOP balance for a month prior to the maximum loan month should have a record for the maximum loan month. For example, if the maximum loan month on file is February 2015, and the loan is current in January 2015 but does not have a February 2015 record, a record will need to be created.
5. Loan Month Issue Date Consistency: Each loan ID should have a record where the loan month equals the issue date. Further, a record must exist for each loan between the issue date and the current file date, charge off date, or fully paid date, whichever is earlier.
6. Loan Age/Days Past Due: Days Past Due value should never be negative. The expected loan age is calculated as the loan month minus the issue date.
7. Charge Off/Fully Paid Fields:
 - a. All instances where the

$$CO\ Amount \neq 0$$

- should have the charge off flag set to 1.
 - b. All loans that have a status of fully paid should have an end-of-period balance > 0.
 - c. BOP principal minus principal received and charge-off amount should equal to 0.
8. Principal, Interest, and Fee Payments:
 - a. Interest paid for the given loan on a given month should equal between the borrower and the originator



- b. Amount paid should equal the sum of principal, interest, and fees paid
 - c. All principal payments, interest payments, and fee payments should be positive.
9. Field Unchanged through the Loan Life: The following fields should remain unchanged and populated through the loan life: loan purpose, loan interest rate, loan grade, loan term, loan state, original principal, issuance date.
10. Consistency of the Recovery Fields: All recoveries should be positive, and should be recorded at the month the loan charges off.

Lending Club Loan Level Data

1. Lending Club Loan Types Considered: As a starting point for the demonstration of the modeling approach, DROP uses the loan level public data from Lending Club. As such, the loan products considered are fixed rate, fixed term, fully amortizing 36 month and 60 month loans issued by Lending Club.
2. Number and Size of Loans: In all, PeerIQ (2015) uses over 9 million loan months of Lending Club data in constructing the model. The table below contains high level descriptive statistics for select items from the dataset.
3. Descriptive Statistics for LC Data: Source: PeerIQ Research

Field	Mean	Standard Deviation	Minimum	Maximum	First Quartile	Median	Third Quartile
Age (Months on Balance)	10.1	8.9	0.0	60.0	3.0	8.0	15.0
Vintage	February 2013	NA	February 2007	February 2015	April 2012	March 2013	February 2014



Original Principal	\$14,254	\$8,254	\$500	\$35,000	\$8,000	\$12,000	\$20,000
Monthly Gross Income	\$6,066	\$4,599	\$250	\$725,549	\$3,750	\$5,167	\$7,333
Term (Months)	42.7	10.8	36.0	60.0	36.0	36.0	60.0
Coupon	13.7%	4.3%	5.3%	29.0%	10.6%	13.5%	16.3%
FICO Origination	699	31	612	847	677	692	717
DTI (ex-mortgage)	16.6%	7.8%	0.0%	39.0%	11.0%	16.0%	22.0%
Total Borrower Accounts	25	11	1	162	16	23	31
Revolving Utilization Rate	57%	24%	0%	892%	40%	58%	75%
Inquiries in Last 6 Months	0.9	1.2	0.0	33.0	0.0	0.0	1.0
DQ Accounts in Last 2 Years	0.3	0.8	0.0	39.0	0.0	0.0	0.0



Months since Last DQ	34	22	0	188	16	31	50
Months since Last Public Record	76	29	0	129	55	79	102
Total Open Credit Lines	11	5	0	90	8	10	14

4. Period of LC Loan Origination: Overall, the sample used contains 245,243 distinct loans originated between February 2007 and February 2015. Average loan size is a little over \$14,000, varying between \$500 and \$35,000 for Lending Club.
5. Variation among the Underwriting Parameters: There is also considerable variation among other under-writing information, including DTI and revolving debt utilization rates, for example.
6. Defaults/Prepayments by Origination Year: Ultimately the goal is to derive insight into termination events (defaults and prepays) from loan pools, and the table below lists some simple summaries of defaults and prepayments by origination year.
7. LC Prepay and Default Exits: Source: PeerIQ Research.

Origination Year	Origination Volume (\$mm)	Prepays (\$mm)	Cumulative Defaults (\$mm)	Cumulative Defaults to Date (%)
2010	132	26	12	8.82%
2011	262	62	27	10.49%
2012	718	164	75	10.39%



2013	1,982	405	135	6.83%
2014	3,504	387	88	2.51%

8. Peaking of Defaults by Vintage: The table above shows that the defaults in the current cycle have peaked for the 2011 and the 2012 vintages.

Loan Credit Model Implementation

1. Discrete Loan Level Status Codes: The approach to modeling starts with the observation that at a given point in time, a loan has several status codes, or 'state's: current, delinquent, fully prepaid, or charged off.
2. Explicit Transition between Loan States: The key conceptual idea is to model transitions between the various states explicitly. For example, a loan may move from a state of 'current' to '30-day delinquent', or move further down the delinquency queue (e.g., move from 30 day to 60 day delinquent). Alternatively, a loan may 'cure' and move from a status of current to delinquent.
3. Exit from the Transition Graph: The only exits for a loan from this network are a transition to prepay or default, which are of course the end statuses we are most interested in. The graph table below illustrates the transition network.
4. Directed State Transition Network Graph: Source: PeerIQ Research.

State	C	P	D3	D6	D6+	D
C	Y	Y	Y	N	N	Y
P	N	Y	N	N	N	N



D3	Y	Y	Y	Y	N	N
D6	N	N	Y	Y	Y	Y
D6+	N	N	N	Y	Y	Y
D	N	N	N	N	N	Y

5. State Transition Network Graph Annotation: Y's indicate directionality of the possible transitions. Note that prepay and default here are “absorbing” states, i.e., states from which exits are not possible. For modeling purposes, and due to data considerations, the 90-180 day delinquent states have been combined into the D6+ category.
 - a. C => Current
 - b. P => Prepay
 - c. D3 => 30 day Delinquent
 - d. D6 => 60 day Delinquent
 - e. D6+ => 90 and more days Delinquent
 - f. D => Default
6. Cross-State Transition Probabilities: Mathematically the probabilities of these moves can be represented in a transition matrix thereby describing the propensity of the borrower to move from one state to another.
7. State -> State Transition Setup:

$$P_{i,j}(t) = \begin{bmatrix} P_{C,C}(t) & \cdots & P_{C,D}(t) \\ \vdots & \ddots & \vdots \\ P_{6+,C}(t) & \cdots & P_{6+,D}(t) \end{bmatrix}$$

Each entry in the matrix represents the probability of the borrower moving from the row state to the column state in a particular month. For example, $P_{C,C}(t)$ represents the probability of



the borrower moving from current to current (that is, staying current) over month t , and so on.

8. Transition Probabilities from MNL Frameworks: The next step is to decide on the parameterization of the entries in the transition matrix, i.e., how does one model the transition probabilities $P_{i,j}(t)$ (commonly referred to as the ‘roll rate’s?).
9. Transition Matrix in Markov Process: The MNL framework has features prominently in the mortgage modeling literature in modeling prepay and default. Such a modeling setup is known as the *Markov* process, which in practice means that each monthly observation of a loan’s transitions is independent of any prior observations.
10. MNL Formulation in Logit Framework: Under the logit assumption inherent in MNL, the transition probabilities assume the following form:

$$P_{i,j}(t) = \frac{e^{\vec{\beta}_0 + \vec{\beta}_1 \cdot \vec{X}_{i,j}(t)}}{1 + e^{\vec{\beta}_0 + \vec{\beta}_1 \cdot \vec{X}_{i,j}(t)}}$$

11. Loan Variables as Regressor Factors: Here $\vec{X}_{i,j}$ represents the matrix of regressors (or predictors) observed at a particular time (such as consumer credit variables, loan age, cohort, for example, and macro-economic variables such as inherent rates). Note that we are not restricted to using the same set of predictors for every transition.
12. Normalization under the MNL Framework: The parametrization above implies that the probabilities $P_{i,j}(t)$ will lie between zero and one, as they should. Further, since the probabilities across a given row should sum up to one, there will always be a status s for which the probabilities are determined as

$$P_{i,s}(t) = 1 - \sum_j P_{i,j}(t) = \frac{1}{1 + \sum_k e^{\vec{\beta}_0 + \vec{\beta}_1 \cdot \vec{X}_{i,k}(t)}}$$



13. Reduction of the Transition Probabilities: In estimating the model, one does not generally estimate all the probabilities from one state to another, as this would make the models unnecessarily complex, especially if the transitions are rare.
14. Current to Charge-Off Transition: While there are various reasons a loan can theoretically transition from current to charge-off (skipping intermediate statuses such as delinquency), such as due to the death of the borrower, these tend to be rare empirically.
15. Current-to-Default Transition Likelihood: This, it is quite intuitive that the probability of going from a state of current to a state of charge-off in a month should be quiet low. As shown in PeerIQ (2015) this monthly transition rate can be seen to be at most 0.03% for 60 month loans. Therefore, this probability is not estimated.
16. Estimation of the Sparse Matrix: Thus, a sparse matrix, which is a subset of all the possible transitions – is estimated. The grid below gives the transitions that are estimated in the model, and those that are not.
17. Sparse Transition Matrix and Determinants: Source: PeerIQ Research.

States	C	D3	D6	D6+	D	P
C	Y	Y	N	N	N	Y
D3	Y	Y	Y	N	Y	Y
D6	N	Y	Y	Y	N	N
D6+	N	N	Y	Y	Y	N

Clearly the number of transitions to be estimated will vary depending on the originator and the observed frequency.

18. Transition Probabilities Using Separate Models: Since there is no restriction to using the same set of regressors for each probability, it is instructive to think of each of the entries estimated in the sparse matrix as separate models.



DROP Credit Model Selection Methodology

1. Cross Validation Based Model Selection: DROP applies a rigorous out-of-sample based testing procedure to estimate each of the models discussed above. The procedure for the model selection is described as follows.
2. Candidate Variables for Model Selection: DROP has identified a set of candidate variables for selection. Those variables and their data types are listed in the table below.
3. Candidate Variables and their Types: Source: PeerIQ Research.

Variable	Implementation Type
Age (Months Remaining)	Evenly Spaced Linear Splines
Term (36 or 60 months)	Categorical Variable
Interaction Variable (Age Spline * Term)	Linear Spline * Categorical Variable
DTI	Continuous Variable
FICO at Origination	Categorical Variable
Vintage (Origination Year)	Categorical Variable
Seasonality (Month of Year)	Categorical Variable
Original Loan Size Bucket	Categorical Variable
Coupon Stack Bucket	Categorical Variable
Loan Purpose	Categorical Variable
Employment Length	Categorical Variable
Inquiries in the Past 6 Months	Untransformed



Monthly Gross Income	Untransformed
Total Outstanding Accounts	Untransformed
Revolving Credit Utilization	Untransformed
Delinquent Accounts in last 2 Years	Untransformed
Total Open Credit Lines	Untransformed

4. Model Variables from Candidate Regressors: Altogether the combinations of the categorical and the numerical variables generate 86 candidate regressors from which the best choice for each transition probability is optimized. The details of the procedure for model variable selection follow.
5. Likelihood Estimation for Traditional MNL: Variable selection was achieved via a regularized logistic regression procedure. In traditional logistic regression one typically optimizes for the values of $\vec{\beta}_0$ and $\vec{\beta}_1$ in the equation above. Applying maximum likelihood across borrowers, the formulation for the transition probability between states (for e.g., borrowers moving from current to prepay) follows, as shown below.
6. Example Current -> Prepay MLE Setup:

$$\text{Max}[L(\vec{\beta}_0, \vec{\beta}_1)] = \sum_{m=1}^N \log P_{c,p,m}$$

where $P_{c,p,m}$ represents the probability of the m^{th} borrower moving from current to prepay (dropping time subscripts for convenience) with the probabilities given from

$$P_{i,j}(t) = \frac{e^{\vec{\beta}_0 + \vec{\beta}_1 \cdot \vec{X}_{i,j}(t)}}{1 + e^{\vec{\beta}_0 + \vec{\beta}_1 \cdot \vec{X}_{i,j}(t)}}$$



7. Penalty Based Regularized Logistic Regression: Regularized logistic regression is similar to the equation above, but it imposes a penalty on $\vec{\beta}_0$ and $\vec{\beta}_1$ to ensure that the coefficients that are not predictive of the transition probability are not unnecessarily added (and thus eliminated from the regression).
8. Current Prepay Penalized MLE Setup:

$$\text{Max}[L(\vec{\beta}_0, \vec{\beta}_1)] = \sum_{m=1}^N \log P_{c,p,m} + \alpha \sum_{j=1}^p |\beta_j|$$

where β_j represents the set of predictors in $\vec{\beta}_1$, α represents the parameter that determines the strength of the penalty function.

9. Optimal Penalty Loading Selection Algorithm: The penalty loading parameter α is selected using the following step:
 - a. Select a possible range of values for α
 - b. For each value of α in the range above, train the logistic regression model on a training dataset, and test the predictive power of the model on a separate test dataset
 - c. The test data is defined by sampling every third observation to minimize the risk of over-fitting the data
 - d. Choose the value of α which results in the ‘best’ performance on the testing dataset
10. Classifier Performance Using Sample AUC: The AUC is a well-known measure of predicting accuracy of a classification technique (Receiver Operating Characteristic (Wiki)). Thus, the ‘best’ in the algorithm above is decided by looking at that value of α that gives the best out-of-sample AUC, e.g., on classifying borrowers who move from current to prepay in a given month.



Empirical Results – Regressor Contribution Weights

1. C -> P Predictor Loading Strength Order:

Strong Positive	Strong Inverse	Weak/Neutral
a. Age_k0.0 b. Total_accounts c. Inq_6m	a. Revolving_utilization b. Total_open_credit_lines c. Age_k0.0_term_60 d. Age_k18.0 e. Dti_ex_mortgage	a. Coupon_20+ b. Cohort_2014 c. Cohort_2013 d. Fico_750_800 e. Lp_debt_consolidation f. Coupon_15-20 g. Cohort_2015 h. Mo_3 i. Mo_5 j. Mo_2 k. Mo_4 l. Size_15_20 m. Mo_10 n. El_10+years o. El_<1year p. Mo_8 q. Mo_9 r. Mo_11 s. Size_gte20 t. Fico_650_700 u. Lp_small_business v. Coupon_<10 w. Cohort_2011



		x. Cohort_2009 y. Cohort_2008 z. Cohort_2010 aa. El_n/a bb. Mo_12 cc. Term_60 dd. Age_k12.0
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2. C -> D3 Predictor Loading Strength Order:

Strong Positive	Strong Inverse	Weak/Neutral
a. Age_k0.0 b. Inq_6m	a. Age_k6.0 b. Monthly_gross_income c. Age_k12.0	a. Coupon_20+ b. Coupon_15-20 c. Dti_ex_mortgage d. Dq_accounts_past_2_years e. Lp_small_business f. Lp_educational g. Cohort_2008 h. Lp_moving i. El_n/a j. Mo_10 k. Lp_other l. Fico_650_700 m. Mo_7 n. Lp_medical o. El_<1year p. Mo_6 q. Mo_11 r. Size_5_10



		s. Mo_9 t. El_10+years u. Lp_major_purchase v. Lp_debt_consolidation w. Mo_3 x. Size_lte5 y. Cohort_2015 z. Mo_2 aa. Mo_5 bb. Fico_750_800 cc. Lp_wedding dd. Cohort_2014 ee. Total_accounts ff. Fico_800_850 gg. Term_60 hh. Lp_home_improvement ii. Age_k24.0 jj. Cohort_2012 kk. Cohort_2013 ll. Age_k24.0term60 mm. Coupon_<10 nn. Total_open_credit_lines oo. Revolving_utilization
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3. D3 -> D3 Predictor Loading Strength Order:

Strong Positive	Strong Inverse	Weak/Neutral
a. Age_k0.0	a. Cohort_2015	a. Coupon_15-20
b. Dq_accounts_past_2_years	b. Cohort_2014	b. Size_15_20



c. Cohort_2009	c. Age_k24.0term_60	c. Fico_650_700
d. Cohort_2008	d. Cohort_2013	d. El_4years
e. Cohort_2010	e. Size_lte5	e. Mo_7
f. Revolving_utilization	f. Total_open_credit_lines	f. Mo_10
g. Size_gte20	g. Age_k18.0term_60	g. Mo_8
h. Coupon_20+	h. Cohort_2012	h. Fico_750_800
	i. Term_60	i. Mo_12
	j. Size_5_10	j. El_3years
	k. El_7years	k. Mo_9
		l. Mo_5
		m. Coupon_<10
		n. Mo_11
		o. Lp_credit_card
		p. El_5years
		q. El_6years
		r. El_8years

4. D3 -> C Predictor Loading Strength Order:

Strong Positive	Strong Inverse	Weak/Neutral
a. Monthly_gross_income	a. Total_open_credit_lines	a. Revolving_utilization
b. Dq_accounts_past_2_years	b. Cohort_2014	b. Age_k12.0term_60
	c. Cohort_2015	c. Mo_3
	d. Dti_ex_mortgage	d. Mo_4
	e. Cohort_2013	e. Lp_moving
		f. Fico_650_700
		g. Size_lte5
		h. Lp_car
		i. Cohort_2008



		j. Lp_medical k. Cohort_2010 l. Size_5_10 m. Coupon_<10 n. Size_gte20 o. Cohort_2009 p. Lp_other q. Age_k12.0 r. Lp_debt_consolidation s. El_7years t. Mo_5 u. El_5years v. El_8years w. Size_15_20 x. El_9years y. Mo_6 z. Inq_6m aa. El_10+years bb. Mo_12 cc. Mo_7 dd. Term_60 ee. Cohort_2012 ff. Age_k30.0 gg. Mo_11 hh. El_n/a
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5. D3 -> P Predictor Loading Strength Order:

Strong Positive	Strong Inverse	Weak/Neutral
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a. Total_accounts	a. Age_k30.0	a. Size_gte20
b. Age_k6.0	b. Age_k0.0term_60	b. Mo_10
c. Monthly_gross_income	c. Dti_ex_mortgage	c. Lp_credit_card
d. Fico_800_850	d. Lp_car	d. Mo_6
e. Cohort_2009	e. Mo_8	e. Mo_7
f. El_10+years	f. Lp_small_business	f. Cohort_2014
g. El_8years	g. Fico_700_750	g. Cohort_2013
h. El_5years	h. Size_lte5	h. Size_15_20
i. Cohort_2015		i. Cohort_2010
j. El_6years		j. Term_60

6. D3 -> D Predictor Loading Strength Order:

Strong Positive	Strong Inverse	Weak/Neutral
a. Age_k0.0	a. Coupon_15_20	a. Size_5_10
b. Total_accounts	b. El_3years	b. Cohort_2012
c. Age_k6.0	c. Coupon_20+	c. Mo_5
d. Dti_ex_mortgage		d. Mo_6
e. Size_lte5		e. El_7years
f. Cohort_2013		f. Fico_700_750
g. Cohort_2014		g. Term_60
h. El_10+years		h. El_<1year
		i. Mo_12
		j. Mo_2
		k. Size_15_20
		l. Size_gte20
		m. El_4years

7. D6 -> D6 Predictor Loading Strength Order:



Strong Positive	Strong Inverse	Weak/Neutral
a. Age_k0.0 b. Age_k6.0 c. Revolving_utilization d. Coupon_20+ e. Total_accounts f. El_4years	a. Cohort_2014 b. Cohort_2015 c. Cohort_2013 d. Cohort_2012 e. Cohort_2011 f. Age_k24.0term_60 g. Age_k30.0term_60 h. Size_lte5 i. Lp_major_purchase j. Term_60	a. Cohort_2009 b. Coupon_15-20 c. El_<1year d. Fico_750_800 e. Mo_5 f. Size_gte20 g. Lp_home_improvement h. Mo_2 i. Lp_other j. Lp_debt_consolidation k. Mo_3 l. Mo_4 m. El_10+years n. Age_k18.0term_60 o. Fico_700_750 p. Lp_credit_card q. Mo_10 r. Mo_12 s. Mo_11 t. Size_15_20 u. El_7years v. El_9years w. Mo_7 x. El_6years y. Size_5_10 z. El_8years aa. Dti_ex_mortgage



8. D6 -> D6+ Predictor Loading Strength Order:

Strong Positive	Strong Inverse	Weak/Neutral
a. Age_k30.0term_60 b. Age_k24.0term_60 c. Age_k18.0term_60 d. Total_open_credit_lines	a. Age_k30.0 b. Total_accounts c. Age_k0.0 d. Cohort_2008 e. Age_k6.0	a. Cohort_2012 b. Coupon_20+ c. Mo_11 d. El_9years e. Cohort_2015 f. Size_gte20 g. Mo_7 h. Fico_650_700 i. El_4years j. Cohort_2013 k. Mo_8 l. Lp_credit_card m. Age_k36.0term_60 n. Mo_12 o. Mo_2 p. Coupon_<10 q. Monthly_gross_income r. El_<1year s. Mo_6 t. Fico_750_800 u. Mo_9 v. Term_60 w. El_10+years x. Cohort_2009 y. El_n/a z. Cohort_2010 aa. Mo_4



		bb. Age_k24.0 cc. Size_5_10 dd. Revolving_utilization ee. Size_lte5 ff. Mo_3
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9. D6+ -> D6+ Predictor Loading Strength Order:

Strong Positive	Strong Inverse	Weak/Neutral
a. Age_k12.0 b. Cohort_2008 c. Cohort_2009	a. Age_k0.0 b. Cohort_2013 c. Cohort_2014 d. Cohort_2012 e. Cohort_2011	a. Mo_11 b. Mo_9 c. Revolving_utilization d. Coupon_20+ e. Fico_650_700 f. Size_gte20 g. Mo_7 h. Coupon_15-20 i. Age_k18.0 j. Mo_12 k. Lp_debt_consolidation l. El_3years m. El_6years n. Mo_6 o. Size_5_10 p. Lp_other q. Mo_4 r. Mo_10 s. Mo_3 t. Size_lte5



		u. Age_k0.0term_60
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10. D6+ -> D Predictor Loading Strength Order:

Strong Positive	Strong Inverse	Weak/Neutral
a. Cohort_2012 b. Cohort_2013 c. Cohort_2014 d. Mo_7 e. Term_60 f. Age_k24.0term_60 g. Mo_12 h. Mo_6 i. Cohort_2011	a. Age_k6.0 b. Age_k12.0 c. Dq_accounts_past_2_years d. Cohort_2008 e. Cohort_2009 f. Total_accounts g. Mo_2	a. Lp_car b. Size_15_20 c. Dti_ex_mortgage d. Mo_10 e. El_6years f. El_9years g. El_n/a h. Mo_5 i. El_5years j. El_7years k. El_3years l. Coupon_15_20 m. Lp_debt_consolidation n. Mo_9 o. El_10+years p. Mo_8 q. Mo_3 r. Size_lte5 s. Coupon_<10 t. Cohort_2010

Empirical Analysis of Seasoning Effects



1. Default and Prepay over Time: Perhaps one of the most important predictors of a loan's prepayment and default rate is just the passage of time. The empirical exhibits of the conditional prepayment C-P probabilities over age (months on balance) by PeerIQ (2015) illustrate that monthly prepayment probabilities generally increase with loan seasoning.
2. Time Impact on the Delinquency: Conversely, monthly rolls to delinquency (and ultimately to default) generally fall as a loan seasons. Conditional on the fact that a loan has not defaulted to a certain point in time, it is more generally likely to prepay and not default in the future.
3. Actual vs. Fitted C-P Probabilities: PeerIQ (2015) find that the fitted C-P probabilities correspond quite well to the actual C-P probabilities indicating that the age splines are quite explanatory (however less so at later ages where there is some noise due to the thinning of the sample size). Nonetheless the general uptrend of prepayment by loan age is evident in both the actual and the fitted C-P probabilities.
4. Hump in the Delinquency Curve: Conversely, rolls into delinquency peak early in the life of a loan (within the first 15 months of age), followed by a gradual decline through the life of the loan. The result is particularly true for the 60 month loans which (due to their longer term) season meaningfully towards the back end.

Analysis of the Vintage/Cohort Effects

1. Vintage/Cohort Impact on Loan: Vintage and cohort effects are designed to capture the difference in prepayment and default experience of loans identical in all respects except for the fact that they were originated at different times.
2. Impact on Prepay/Delinquency Rolls: While the monotonicity of the prepayment rates are likely due to the seasoning effects described above (for e.g., the relative lower prepayment probability of the younger 2015 vintage loans), the peak in fitted delinquency rolls around the 2012 vintage for the 60 month loans and the 2013 vintage for the 36 month loans is both interesting and consistent with the prepay/default exist in LC data seen before.



Analysis of Empirical Seasonality Effects

1. Usage of Origination DTI Metric: Once again the DTI metrics used in the model are at origination. Again, the results are quite intuitive and fit the data well.
2. DTI Impact on Prepay/Delinquency: Borrowers with higher DTI tend to prepay slower and tend to roll to delinquency at a higher rate, albeit not with the same effect as that observed on prepayment.

CPR And CDR Curve Estimation

1. Definition of CPR and CDR: The fit probabilities (and therefore the sparse transition matrices) obtained from the above are used to derive CPR and CDR curves. CPR and CDR curves are defined as

$$CDR(t) = 1 - \left[1 - \frac{Default(t)}{BeginBalance(t)} \right]^{12}$$

and

$$CPR(t) = 1 - \left[1 - \frac{Prepay(t)}{BeginBalance(t)} \right]^{12}$$

2. CPR/CDR for Loan Pools: CPR and CDR therefore represent an annualized measure of prepay and default for every dollar of principal outstanding at the start of a period. Clearly, in order to estimate the CPR and CDR, we need to project the cash flows for a given pool of loans.



3. Projecting Each Loan in Pool: A first step to projecting the cash flows is to project the status of each loan in the pool at all future points in time. This is achieved by using the transition matrix

$$P_{i,j}(t) = \begin{bmatrix} P_{c,c}(t) & \cdots & P_{c,d}(t) \\ \vdots & \ddots & \vdots \\ P_{6+,c}(t) & \cdots & P_{6+,d}(t) \end{bmatrix}$$

4. Using MNL Loan Transition Probabilities: Probabilities are estimated at all points in time, as per the earlier section. Suppose, for example, that at a given time t the loan status is current. Aw: The multinomial distribution for where the loan can transition to at time $t + 1$ is used, and it is simply by the first row of the transition matrix.
5. Loan Target Status Random Draw: Therefore, a random draw is made from this distribution to project the status of the loan at time $t + 1$. If, in that draw, the loan ended up in a status of 30-day delinquent, a simulation is done using the second row of $P_{i,j}(t + 1)$ and so on, until the earliest of one of maturity, default, or prepay.
6. Projection of Loan Cash Flows: After having obtained the future status of all the loans in the pool, we can compute the cash flows appropriate to the product under consideration (in this case the 36 month vs. the 60 month fixed rate loans).
7. Loan State Dependent Cash Flow: For example, in the case of Lending Club, one can continue to apply the monthly fixed payment on the loan if the loan is in current status, and accrue interest if the loan becomes delinquent, or discontinue further payments altogether if the loan voluntarily prepays or defaults.
8. CPR/CDR from Cash Flows: Such logic allows the computation of the loan balances from period to period, and ultimately the CPR and the CDR above.
9. CPR and CDR Simulation: As a final example, we project the loan status, derive cash flows, and compute CPR and CDR for the entire population of outstanding Lending Club loans – outstanding as of January 2013 – and examine the results. DROP generates the results of one simulation on the portfolio containing 14,000 Lending Club loans as at 1 January 2013.
10. Actual vs. Simulated CPR/CDR: As evidenced from the above, the projection for the CPR and the CDR agrees with the realized values for much of the projection period. For CDR,



there is some volatility towards the latter end of the projection period where the sample thins out, and where the number of defaults (as a rare transition) can have a meaningful impact.

11. Multi-Path CPR/CDR Simulation: In addition to such ‘single path’ projections of CPR and CDR on a portfolio, because the calibration produces a *distribution* of transitions, one can generate multiple paths for the CPR and CDR on a loan portfolio. Using the model DROP simulated 10 paths for the LC portfolio resulting in a distribution of CPR and CDR for the portfolio.

Credit Model Future Enhancements

1. Inclusion of Macro-economic Regressors: For the purposes of keeping the analysis simple, DROP’s model excludes macro-economic regressors from the analysis. However economic variables can be integrated quite easily into the structure of the model.
2. Stochastic Simulation of Market Variables: In future versions we plan to add in carefully selected market variables such as interest rates or unemployment, which can be used as scenario variables, or stochastically simulated via parameters calibrated from the prices of traded instruments (e.g., interest rate options).

References

- PeerIQ (2015): PeerIQ Analytics Credit Model Methodology, Version 2.0.
- [Receiver Operating Characteristic \(Wiki\)](#).



Asset-Backed Requirements

All Asset-Backed Desks

1. Consolidated View of Books/Inventories: Ability to provide a consolidated view of trader positions/risk/PnL for all books/inventories managed by any given ABF trader.
2. Cash, Derivative, and Loan Products: This view should be capable of including all USD and non-USD cash, derivative, and loan products, and has capability to show opening day positions as well as isolate trading activities for the day.
3. Creation of a Bespoke View: As an example, a trader should have the capability to create a view that best fits their business. All bespoke views must be exportable to excel.
4. Examples of Trader View Requirements: These are listed below, but they are not exhaustive.
5. Position Blotter:
 - a. TSY Positions
 - b. TBA Positions
 - c. CMO Positions
 - d. MBS Positions
 - e. Swaps Positions
6. Position Blotter – Super Portfolio View: To group specific books/inventories.
7. Position Blotter - Desk View: Desk view to present the overall desk.
8. Trade Blotters:
 - a. Cash Trading
 - b. Derivative Trading
9. Summary Views: The below calculations are to be computed at the instrument level.

Summary views, i.e., pivot tables, will summarize this information:

 - a. PnL
 - b. Attributions
 - c. Bucketed Risk



- d. 10Y Hedge Equivalents

Search, Filter, Group, and Sort Capability

1. Searching for Strand of Characters: The user shall have the ability to search in consolidated trade view by entering a specific ticker/CUSIP or any strand of characters into a search bar.
2. Trades with Specific Search Criteria: The result should be a view of only trades which include the specific criteria entered in search field.
3. Example:
 - a. Enter “COMET 2014” into search bar
 - b. All trades including combination of character “COMET 2014” shall populate in either trading or position blotter.
 - c. User shall have the ability to extract the data into excel format.

Position Management

1. Functionality of the Position Blotter: The position blotter should provide the ability to input cash flow assumptions, prepay speeds, cash flows – BBCF, cash flow timing – to call, trader marks, hedge ratios, and pay-ups.
2. Accuracy of the Positions Displayed: The position MUST always be accurate and account for trades booked in other systems. The consolidated view will be the gold source for position management.
3. Creating Custom Product Position Blotters: Ability to create separate position blotters for Repo/Facility/External Pricing.

Position Summary Reports



1. Pivot Table for Position Summary: Pivot table concepts that summarize the position, e.g., CMBS Risk Report, NACMO risk report – and UI should be able to support multiple instances, i.e., 10 multiple tables.
2. Limits for Market Risk Mandate: Trader should have the ability to utilize the limits specified in the Market Risk Mandate – either an electronic source or a manual input – to measure the appropriate risk metric against the limit.
3. Real time Updates for Controls: Should be real-time and update as position updates. Includes the ability to view reports on DV01 limit, CR01 limit, Balance Sheet limit, Position Aging, etc.

Historical

1. Snap of the Historical Metrics: System should have the capability to snap historical views of Positions, Risk, PnL, etc., and that data is available for future analysis.
2. Capture End-of-Day Metrics: The snap view would capture the end-of-day positions, risk, PnL etc.

Pricing

1. Market Data for Separate Curves: Curve data for LIBOR, E/I/J/N curves should be available and at closing.
2. Firmwide Unified Marks Infrastructure: There should be one source for these curves and it should match what is used firm-wide to calculate PnL attribution.
3. Applicability to Positions and Hedges: The curves would be the same and desk/asset positions as well as hedge positions.
4. Applicability to Positions and Hedges: These would be the same curves applied to any desk asset as well as hedge positions.



5. Intex/Bloomberg Cash flow Assumptions: This should have the ability to handle all Intex/Bloomberg cash flow assumptions.
6. Mark Movements to Benchmark Securities: Allow traders to benchmark security mark movements to other securities, as shown in the next three items.
7. Price CUSIP to Generic Categories: Matrix price CUSIPs to generic categories, e.g., CMBS 3.0 AAA
8. Benchmark to other Security Prices: USD, CDX, TBA, etc.
9. Benchmark to Custom User Index: e.g., 80% 2Y UST, 20% FNMA 4.0

PnL

1. PnL and Carry for all Positions
2. PnL for Opening/Intraday: Must segregate PnL for opening position and day one trade.
3. Attribution Analysis of PnL Volcker: Calculation/display of proper Volcker PnL attribution analysis.
4. Volcker PnL Attribution Analysis
5. Real-time PnL on Changes: PnL updates as trades are booked and market information changes.
6. Viewing Live, Close, and EOD: Ability to view PnL based on user specified curve, e.g., Live, Close, EOD.
7. No PnL Re-computation after Marking: Dollar prices are sent to downstream systems and the system does not re-compute dollar prices post daily trader signoff on PnL and risk.

Interest Rate Risk

1. Options on Viewing IR Risk: Provide options on how to view interest rate risk.
2. 10Y/5Y/Other Equivalents
3. Bucketed Risk KRD across Curves: Across Discount curves I/J/N/E/Z/A/LIBOR



4. User to Enter KRD Metrics: Allow user to input KRD's from external sources – e.g., Black Rock Green Package.
5. Risk for Trading and Hedges: Must calculate interest rate risk for ALL trading and hedging positions – cash and derivative, using Risk Mandates for Approved Instruments:
 - a. Model based Computations – par rate analysis
 - b. Hedge ratios
6. Live and On-Demand Risk: Risk updates as trades are booked and markets move.

Offerings

1. Construction of Multiple Offering Sheets: The ability to create multiple offering sheets with different offers and/or offering prices.
2. Using Different Cash-flow Assumptions: User should also be able to use different cash flow assumptions from the mark and the user has the ability to create format for the offering sheet.

What-if Analysis – Scratch Pads

1. Mock up Potential New Trades: Risk scratchpad should also allow user to sketch new buys/sells.
2. Duration Based Neutral Hedging Strategy: Suggest duration neutral hedging strategy based on user specified hedge instruments – UST, Swaps, etc.
3. Execute Hedges and Book Positions: Execute hedge trades and book trading position buys/sells into trade entry system from Scratchpad.
4. PnL Scratchpad for Marks/PnL: PnL scratchpad should allow user to provide segregated marks and PnL screen.
5. Marks Based Prices/PnL: Recompute prices and PnL based on trader what-if marks
6. Save Assumptions/Update Marks from Scratchpad



7. Roll Hedge Scratchpad: The Roll Hedge Scratchpad should allow the user a workspace to specify specific hedge positions to roll into their equivalent positions.
8. Old to New 10Y UST: Example, old 10Y UST to new 10Y UST or current swap ID XXX to a new benchmark 10Y swap.

Non-Agency RMBS/Agency RMBS Pass-through

1. Automated Market Data Feed: An automated market data feed for Treasuries/RBA Closings/High Yield/Investment Grade Indexes and Swaps used for the hedging of desk positions should be incorporated into the desk risk spreadsheet. User should have the ability to specify source of TSY pricing.
2. ABF Desk Automation Functionality: In future state, this capability could provide efficiency in desk processes and be used by any ABF desk regarding the automation functionality.
3. Example of TSY Pricing Sources:
 - a. Wells Treasury Trading
 - b. CDX Spreads
 - c. BBG



CMBS CREF Model

Overview

1. Non-regulatory CMBS CREF Loan Model: The model is used for downstream risk treatment which may include general VaR GVaR, stressed VaR SVaR, and stress scenarios for non-regulatory purposes.
2. Loan Products Covered: The CMBS CREF Loan product to be covered by this model includes conduit loans, non-conduit loans, and MFC loans.

Business Justification

1. Securitizing Loans in CMBS Warehouse: Banks buy CREF loans and houses them in a CREF warehouse. These loans are then subsequently securitized in the CMBS market place and sold to CMBS investors.
2. Pricing/Risking of Securitized Loans: The CMBS CREF Loan Model is used to risk-manage these CREF “warehoused” on the bank’s balance sheet. It handles pricing and risking for CREF loans and agency MFC loans.

Glossary of Terms

1. CMBS: Abbreviation for Commercial Mortgage-backed Securities.
2. CREF: Abbreviation for Commercial Real Estate Finance.
3. GVaR: Abbreviation for General Value-at-Risk GVaR. GVaR can refer to 10-day or 1-day GVaR of 99% confidence if it is not explicitly specified.



4. IO: Abbreviation for Interest Only.
5. LIBOR: Abbreviation for London Interbank Offering Rate.
6. MRS: Abbreviation for Market Risk System – the system at the bank for market risk capital calculations.
7. MFC: Abbreviation for Multi-family Commercial.
8. Non-Reg: Abbreviation for “Non-regulatory”.
9. PnL: Abbreviation for Profit and Loss.
10. Risk Factor: Variables within a pricing function that drive the change of the price. For example, interest rates, volatility, credit spread, mortgage basis etc.
11. SVaR: Abbreviation for stressed VaR. SVaR refers to the 10-day VaR of 99% confidence during the top-of-the-house TOH 1 year stress period.

In-scope Items

1. Price Values and Sensitivities: Prices and sensitivities under various circumstances, as well as the corresponding PnL.
2. Products in Scope for Handling: The products in scope for the migration include conduit loans, non-conduit loans, and MFC loans.
3. Risk Factors: The identified risk factors include CMBS2 AAA, CMBS2 BBB, and CMBSFNMADUS.

Out-of-Scope Items

1. Model Tuning
2. FRTB Internal Models Approach IMA Requirements
3. FRTB Standardized Approach SA Requirements
4. Intraday Reporting



Dependencies

1. Market Data Inputs: The market data inputs, such as LIBOR curves and generic credit curves, are taken from external sources.
2. Position and Contract Inputs: Contract and position parameters are from the MRS database.
3. Spread Solver used by Matching Pricer: A spread solver based on the Newton-Raphson method is used in matching market price.

Assumptions

1. Adequacy of Delta-Gamma PnL: For the risk calculation, the current model includes all the key factors with sufficient granularity and the Delta-Gamma approximation is sufficient for the PnL calculations for the management VaR, even under stress market conditions.
2. CMBS CREF Model Uses Mock Securitization: The CMBS CREF model prices the loans by the Mock Securitization approach, in which only AAA and BBB tranches are considered.
3. Prepayment for CMBS CREF Loans: Zero prepayment is assumed for entire loan term for CMBS CREF loans.
4. Auto, Commercial, and Corporate Loans: The prepayment speed is zero for auto loans, which are short terms, and commercial and corporate loans, which have a high prepay penalty.
5. HELs, HELOCs, and NPLs: For HELs, HELOCs, and NPLs, the prepayment speed is assumed constant for whole loan term.
6. Rating Migration and Default Risk: No event risk of rating migration and default risk is assumed for loan borrowers and the cashflows of loans are determined solely by loan term and coupon rates.
7. No Specific Market Data Calibration: Also, the models are used directly for pricing and risk sensitivity calculation with an assigned generic discounting curve, without calibration.



Data Inputs

1. Rating-Based Loan Reference Curve: Most of the generic rating-based loan curves are sourced from the third-party vendors such as Bloomberg and Reuters, with the remaining collected from the bank's trading desk/internal research group.
2. Rating-Based Loan Forecast Curve: Forecast curves are for floating deals only, generally 3M LIBOR.
3. Issuer appropriate Loan Credit Curve: The issuer-appropriate credit curves are generic issuer type and rating-based yield curve built from PAR rates.
4. CMBS2 AAA/BBB, and CMBSFNMADUS: CMBS2 AAA and BBB curves for non-agency CMBS CREF loans are used, and CMBSFNMADUS curve for agency MFC loans.
5. Input Bond Market Prices: Market price quotes typically come from the Tier 2 database in MRS if they are available. They are used for price matching.
6. Components of the Deal: The deal information includes principal, interests, and balance schedules, credit rating, etc.
7. Extraction from the MRS Database: These inputs are typically fed from the corresponding staging tables in the MRS database and are sourced from the various source systems.
8. Maturity of the CREF Loan:
9. CREF Loan First Payment Date:
10. CREF Loan Original Balance:
11. CREF Loan Treasury Settings: Original treasury rate and coupon spread.
12. CREF Loan Service Fee:
13. CREF Loan IO Only Term: If not available, set to zero.
14. CREF Loan Payment Frequency: If not available, set to Monthly.
15. CREF Loan Reset Frequency: If not available, set to Payment Frequency.
16. CREF Loan Payment Type: Fixed or float.
17. Loan Prepayment Type and Speed: Pricing assumptions such as prepayment type and prepayment speed are specified by the trading desks, with a default value set to zero.



Processing Requirements

1. CREF Warehouse Loan Amortization Scheme: CREF Warehouse loan model is based on simple amortization schedule. Currently all CREF loans are fixed-rate loans with monthly/quarterly amortization. Yet the model has been extended for floating-rate CREF loans.
2. Principal and Interest Payments for Fixed Rate Loans: Principal and interest payment for interest-rate loans is calculated using the formula:

$$\text{Monthly Rate} = \frac{\text{Original Treasury Rate} + \text{Loan Spread}}{12}$$

$$\text{PI Payment} = \frac{\text{Original Balance} \times \text{Monthly Rate}}{1 - \frac{1}{(1 + \text{Monthly Rate})^{\text{Original Amortization Term}}}}$$

3. Cashflows for Fixed-Rate Loans: Cashflow for fixed-rate loans is calculated as below.
4. Fixed-Rate Loans Non-Interest-Only IO Period:

$$\text{Cash Flow} = \text{PI Payment}$$

5. Fixed-Rate Loans IO Period:

$$\text{Cash Flow} = \text{IO Amount} + \text{Service Fee Amount}$$

where

$$\begin{aligned} \text{IO Amount} &= \text{Current Balance} \times \text{Coupon Year Fraction} \\ &\quad \times (\text{Coupon Rate} + \text{Service Fee}) - \text{Service Fee Amount} \end{aligned}$$

6. Fixed-Rate Loans Final Period:



$$\text{Cash Flow} = \text{PI Payment} + \text{Service Fee Amount} + \text{Current Balance}$$

7. Principal and Interest Payment for Floating-Rate Loans: Principal and interest payment for floating-rate loans in coupon period i is calculated using the formula:

$$\text{PI Payment}(i) = 1 - \frac{1}{[1 + \text{CPI}(i)]^{\text{Remaining Term}}}$$

8. Coupon Schedule i : Based on a given floating index, coupon reset frequency code, and spread.
9. CPN(i): The coupon period for period i according to the coupon schedule.
10. Current Balance: The remaining balance.
11. Cash Flow for Floating-Rate Loans: Cashflow for floating-rate loans in coupon period i is calculated as shown below.
12. Floating-Rate Loan Non-IO Period:

$$\text{Cash Flow}(i) = \text{PI Payment}(i)$$

13. Floating-Rate Loan IO Period:

$$\text{Cash Flow}(i) = \text{IO Amount}(i) + \text{Service Fee Amount}$$

where

$$\text{IO Amount}(i)$$

$$\begin{aligned} &= \text{Current Balance} \times \text{Coupon Year Fraction} \times \text{CPN}(i) \\ &\quad - \text{Service Fee Amount} \end{aligned}$$

$$\text{Service Fee Amount} = \text{Current Balance} \times \text{Service Year Fraction} \times \text{Service Fee}$$



14. Floating-Rate Loan Final Term:

$$Cash\ Flow(i) = PI\ Payment(i) + Service\ Fee\ Amount + Current\ Balance$$

15. Loan Price Calculation: Loan price is calculated using the formula

$$Price = \sum_{i=1}^N D_f(i) \times CF(i)$$

where $D_f(i)$ is the discount factor of the CMBS curve at period i , and $CF(i)$ is the cashflow at period i . Cashflows are based on 100.0 notional.

Risk Factor Sensitivity Requirements

1. First and Second Order Sensitivities: Under the typical bank risk framework, the daily calculations of first and second order sensitivities of several risk factors are required.
2. VaR Types and their Horizons: These sensitivities will be applied with the deal-gamma VaR methodology in the historical simulation to calculate the downstream 1-day GVaR, 10-day GVaR, and 10-day SVaR. The risk factors and their identified sensitivities are detailed in the following requirements.
3. Risk Factors and their Sensitivities: Delta and gamma values are calculated under shifts to the forecast curve – usually 3M LIBOR – and discount curves – CMBS2 AAA and BBB curves for non-agency loans, and CMBSFN MADUS curve for MFC loans – with the formula below: $Delta = \frac{PV_{up} - PV_{down}}{2 \times shift}$ $Gamma = \frac{PV_{up} + PV_{down} - 2 \times PV}{shift^2}$
4. Settlement Convention for Calculating Sensitivities: It is noted that the $T + 0$ settlement convention should be used in calculating sensitivities.



5. Settlement Conventions for Calculating OAS: This applies to the OAS calculations as well since the price from ICE is also based on $T + 0$ settlement convention.
6. Curve Building Analytics Function: For CREF loan model the following analytics functions are used:

$$\text{Floating Rate} = \text{fixFwd}(\text{forecast_curve}, \text{startDate}, \text{endDate}, \text{act}/360, \text{"mth"})$$

$$\text{Discount Rate} = \text{FixDf}(\text{disc_curve}, \text{date})$$

7. Discount Curve Mapping: The discount curve is mapped by product type and issue rating.

Process Outputs

1. Sensitivity Tenor PnL Attribution Outputs: Final output tenors are based on the final PnL attribution outputs and are subject to change as the need arises for analysis.
2. Final Output #1 - Base NPV: For each trade, the Base NPV is the NPV for this trade under current model conditions.
3. Record of Valuation Output Details: Certain valuation details such as model used, solver spread, some key inputs, warnings, and errors are written into a database table.
4. Final Output #2 - Shifted NPV: For general VaR, up and down NPV's are calculated with the system-defined shift size for each risk driver.
5. Baseline Scenario NPV Downstream Processing: These three NPVs and the shift size are fed into downstream MRS process to calculate the deltas and gammas to the corresponding risk driver.
6. Stresses Scenario NPV Downstream Processing: For stress scenario valuation, all the drivers are shifted according to the specified shift size and are simultaneously put into the model to re-evaluate.
7. Recording of the Scenario PnL: The new NPV and the change in NPV will be saved into a database as the scenario PV and PnL.



8. Final Output #3 - PnL: PnL can be calculated either from NPV change or using the Delta-Gamma approach with the sensitivities and shifts from MRS. The results should be the same.

Portfolio Composition

1. Purpose of the CREF Model: CMBS CREF model is used to risk-manage these commercial real-estate loans “warehoused” on the bank’s balance sheet.
2. Residence Duration of the Loans: Generally, these loans stay on the bank’s books for a typical period of 90 days or less. Once securitized and sold out, they are no longer on the books.
3. Risk carried by the Loans: However, in the interim, these loans carry interest rate risk and CMBS spread risk to the bank.
4. Pool Growth During the Securitization: The risk comes from the necessary time lag for a pool to grow to sufficient size for the next securitization and the fluctuation of the mortgage rates over the period.
5. CREF Loan Type #1: Interest-only Loan.
6. CREF Loan Type #2: Principal and interest loan.
7. CREF Loan Type #3: Mixed loans that pay interest only for some term and the start amortizing.
8. Liquidity of the Individual Loans: Because individual commercial real-estate loans can sometimes be illiquid and ratings information can be difficult to obtain, a simple securitization model is used to evaluate the risk of the loans.
9. Components of the Eventual Securitization: The CMBS CREF Loan Model assumes that the loans can be priced and viewed, for risk components, as components of a target upcoming CMBS securitization.
10. Inventories that Feed into Model: The loans for the CMBS CREF Loan Model are fed from the following inventories.
11. Conduit: All conduit CREF loans are fixed-rate loans.



12. Non-conduit: The CREF loans from this inventory are sometimes called “large loan”. They are fixed-rate or floating rate.
13. MFC Loans: Multi-family loan trades are processed per market risk request since most are agency loans. The loans can be fixed-rate or floating rate.

Weights for MFC Loans

1. MFC CREF Loan Processing Rationale: MFC loans are processed by the CREF Loan Model with one driver – CMBSFNADUS – per Market Risk request because most of them are agency loans.
2. Risk Drivers for Conduit/Non-conduit: The valuation of conduit and non-conduit CMBS CREF loans have two risk drivers – CMBS2 AAA and CMBS2 BBB.
3. Rationale for AAA and BBB: Only two ratings AAA and BBB are used for the following reasons.
4. Largest Components of CMBS Issuance: Ratings AAA and BBB tend to be the largest components of CMBS Issuance.
5. Series Availability for other Ratings: Historical time series for other CMBS ratings tend to be sporadic and non-homogenous due to limited consistent information.
6. Weight of the Risk Drivers: The weight for the AAA and the BBB risk drivers are set differently for conduit CREF loans than for non-conduit CREF loans.
7. Risk Drivers for Conduit Loans: For conduit CREF loans, 80% of the risk on the loan pool is allocated to an AAA time series, while 20% is mapped to a BBB time series.
8. Risk Driver for Non-conduit Loans: The weights are 48% and 52%, respectively, for AAA and BBB time series for non-conduit CREF loans.
9. Risk Driver for MFC Loans: For MFC loans, CMBSFNADUS driver is used with 100% weighting.

Mark-to-Market Matching



1. Forward Spread using One-dimensional Solvers: In order to match the bond market price, each position has its own forward spreads that are solved by a spread solver.
2. Mark-to-market Matching Process: A mark-to-market matching process is involved in the pricing. The process applies the Newton-Raphson method to find the solver spread that achieves perfect matching of modeled and observed price for each position

$$\text{Threshold} = 1 \text{ cent}$$

3. Adjusted Forward Curve Applied to Geeks: The forward curve after adjusting for the spread solver serves as the basis for the calculation of Greeks for the VaR estimates.
4. Failure of the Spread Solver: One limitation of the spread solver could be that a hard-coded limit could be reached, and this may fail the trade. On general, however, when a spread solver fails, it is either due to wrong static data input or due to wrong market place.

Technical Details

1. Baseline Fixed-Rate CREF Loan #1: $PI \text{ Payment} = \frac{\text{Original Balance} \times \text{Loan Rate}}{1 - (1 + \text{Loan Rate})^{-\text{Original Amortization Term}}}$
2. Baseline Fixed-Rate CREF Loan #2: $\text{Loan Rate} = \frac{\text{Gross Coupon} + \text{Spread}}{12}$
3. Baseline Fixed-Rate CREF Loan #3: $\text{Coupon Rate} = \text{Gross Coupon} - \text{Servicing}$
4. Baseline Fixed-Rate CREF Loan #4: $\text{Accrued Interest} = \text{Current Scaled Balance} \times \text{Accrued Interest Year Fraction} \times \text{Coupon Rate}$
5. Baseline Fixed-Rate CREF Loan #5: $\text{Current Scaled Balance} = \text{Remaining Balance} -$
initially set to 100.
6. Baseline Fixed-Rate CREF Loan #6: $\text{Service Fee Amount} = \text{Current Scaled Balance} \times \text{Service Year Fraction} \times \text{Service Fee}$
7. Baseline Fixed-Rate CREF Loan #7: $\text{IO Amount} = \text{Current Scaled Balance} \times \text{Coupon Year Fraction} \times (\text{Coupon Rate} + \text{Service Fee}) - \text{Service Fee Amount}$



8. Baseline Fixed-Rate CREF Loan #8: Service Year Fraction =

$$\text{YearFrac} (\text{Previous Payment Date}, \text{Next Payment Date}, \text{Servicing Basis})$$
9. Baseline Fixed-Rate CREF Loan #9: Coupon Payment Year Fraction =

$$\text{YearFrac} (\text{Previous Payment Date}, \text{Next Payment Date}, \text{Coupon Basis})$$
10. Baseline Fixed-Rate CREF Loan #10: Servicing Basis = Coupon Basis = Basis =

$$\text{ACT}_{360}$$
11. IO Period Fixed-Rate CREF Loan #1: Cash Flow = IO Amount + Service Fee Amount
12. IO Period Fixed-Rate CREF Loan #2: Principal = 0
13. IO Period Fixed-Rate CREF Loan #3: Remaining Balance = Current Scaled Balance
14. Fixed-Rate Principal + IO Period #1: Cash Flow = $\frac{\text{PI Payment} \times 100}{\text{Original Balance}}$ scaled to 100 par.
15. Fixed-Rate Principal + IO Period #2: Principal = Cash Flow – IO Amount –
 Servicing Fee Amount
16. Fixed-Rate Principal + IO Period #3: Remaining Balance = Current Scaled Balance –
 Principal
17. Fixed-Rate Loan Final Term #1: Principal = Current Scaled Balance
18. Fixed-Rate Loan Final Term #2: Remaining Balance = Current Scaled Balance –
 Principal
19. Fixed-Rate Loan Final Term #3: Cash Flow = IO Amount + Service Fee Amount +
 Principal
20. Baseline Floating-Rate CREF Loan #1: PI Payment =

$$\frac{\text{Current Scaled Balance} \times \text{Loan Rate}}{1 - (1 + \text{Loan Rate})^{-\text{Remaining Amortization Term}}}$$
21. Baseline Floating-Rate CREF Loan #2: Loan Rate = $\frac{\text{Floating Rate} + \text{Margin}}{12}$
22. Baseline Floating-Rate CREF Loan #3: Coupon Rate = Floating Rate + Margin –
 Servicing
23. Floating-Rate Period Cash Flow #1: Remaining Amortization Term =

$$\text{Original Amortization Term} + (\text{Remaining Term} - \text{Term}) - (i - 1)$$
24. Floating-Rate Period Cash Flow #2: Accrued Interest = Current Scaled Balance ×
 Accrued Interest Year Fraction × Coupon Rate



25. Floating-Rate Period Cash Flow #3: $\text{Current Scaled Balance} = \text{Remaining Balance}$
initially set to 100
26. Floating-Rate Period Cash Flow #4: $\text{Service Fee Amount} = \text{Current Scaled Balance} \times$
 $\text{Service Year Fraction} \times \text{Service Fee}$
27. Floating-Rate Period Cash Flow #5: $\text{IO Amount} = \text{Current Scaled Balance} \times$
 $\text{Coupon Year Fraction} \times (\text{Coupon Rate} + \text{Service Fee}) - \text{Service Fee Amount}$
28. Floating-Rate Period Cash Flow #6: $\text{Service Year Fraction} =$
 $\text{YearFrac}(\text{Previous Payment Date}, \text{Next Payment Date}, \text{Servicing Basis})$
29. Floating-Rate Period Cash Flow #7: $\text{Coupon Payment Year Fraction} =$
 $\text{YearFrac}(\text{Previous Payment Date}, \text{Next Payment Date}, \text{Coupon Basis})$
30. Floating-Rate Period Cash Flow #8: $\text{Servicing Basis} = \text{Coupon Basis} = \text{Basis} =$
 ACT_360
31. Floating-Rate IO Period #1: $\text{Cash Flow} = \text{IO Amount} + \text{Service Fee Amount}$
32. Floating-Rate IO Period #2: $\text{Principal} = 0$
33. Floating-Rate IO Period #3: $\text{Remaining Balance} = \text{Current Scaled Balance}$
34. Floating-Rate Fixed Principal Amortization #1: $\text{Cash Flow} =$
 $\text{Fixed Principal Amortization Amount} + \text{IO Amount}$
35. Floating-Rate Fixed Principal Amortization #2: $\text{Principal} =$
 $\text{Fixed Principal Amortization Amount}$
36. Floating-Rate Fixed Principal Amortization #3: $\text{Remaining Balance} =$
 $\text{Current Scaled Balance} - \text{Principal}$
37. Floating-Rate Principal + IO Period #1: $\text{Cash Flow} = \text{PI Payment}$
38. Floating-Rate Principal + IO Period #2: $\text{Principal} = \text{Cash Flow} - \text{IO Amount} -$
 $\text{Service Fee Amount}$
39. Floating-Rate Final Term #1: $\text{Principal} = \text{Current Scaled Balance}$
40. Floating-Rate Final Term #2: $\text{Remaining Balance} = \text{Current Scaled Balance} -$
 Principal
41. Floating-Rate Final Term #3: $\text{Cash Flow} = \text{IO Amount} + \text{Service Fee Amount} +$
 Principal



- 42. Loan Ratings Cashflow Partition #1: For conduit loans, cashflows are partitioned 20% to BBB and 80% to AAA and discounted accordingly.
- 43. Loan Ratings Cashflow Partition #2: For non-conduit loans, cashflows are partitioned 52% to BBB and 48% to AAA and discounted accordingly.
- 44. Loan Ratings Cashflow Partition #3: For MFC loans, 100% CMBSFN MADUS driver is used instead.



US Agency MBS Pricing Model

Summary

1. Focus of this Chapter: The objective for this chapter is to come up with the risk metrics for Agency MBS trading.
2. OAS Based Risk Framework: The framework for the risk metrics is based on the industry standard of Option-adjusted-spread – OAS – calculation method.
3. Calculating the OAS: This method takes the expected value – the mean NPV – across the range of all possible rate scenarios when discounting each scenario's actual cashflows with the benchmark yield curve plus a spread equating to the market price of the mortgage bond – observed price. This spread is referred to as the OAS (Hayre (2001)).
4. Role of PolyPaths: PolyPaths does the simulation and calculation of the cashflows and computes the OAS.
5. Use of Monte-Carlo: Monte-Carlo simulation is used to produce a set of risk-neutral interest rate paths and thus calculate the expected value and the OAS.
6. Path-wise Sensitivities: Cashflow is then generated for each path. Through this process, different risk metrics can also be computed that contribute to pricing change, such as DV01.
7. Underlying Interest Rate Model: The underlying methodology is based on the PolyPaths' Displaced Diffusion and the Normal LIBOR Market Model. PolyPaths uses normal volatilities to generate the normal vega, which feeds into VaR.
8. Lognormal and CEV Skew LMM: Previously, the PolyPaths' Lognormal and the CEV skew LIBOR Market Model allowed negative interest rates.
9. Intrinsic Zero-Rate Boundary: However, with either the absorption boundary for CEV Skew model or the intrinsic property of a lognormal interest rate distribution, forward rates are not allowed to cross the zero-rate boundary.
10. Initial Negative Rates: Thus, negative interest rates are not simulated unless the initial yield curve already contains negative rates.



11. Closed-form for Normal/Displaced Diffusion Models: Note that both the Normal Model and the Displaced Diffusion Models have their corresponding closed-form option pricing formulas, similar to the Black formula: hence, calibration can be carried out like other LIBOR Market Models.
12. Volatility of the Chosen Models: More importantly, the option formulas for the Normal Model and the Displaced Diffusion Model are well-defined even if the swaption's ATM strike is negative. There is not mathematical difficulty for interest rates to enter negative territory for either model.

Product/Portfolio Overview

1. Scope of US Agency MBS Trading: The scope of US Agency MBS Trading is limited to TBA and specified pools only in this chapter – Fannie, Freddie, and Ginnie Mae backed residential mortgage-backed securities.
2. PnL Explain Components: If the risk metrics defined by the Market Risk Policy are Delta, Vega, Current Coupon Spread, and OAS, then the model PnL should be explained by these risk metrics:

$$PnL = PnL(Rates) + PnL(Vega) + PnL(Credit) + PnL(OAS)$$

Model Requirements

1. Requirement #1: At least 256 paths are required to ensure the convergence of the Monte-Carlo simulation.
2. Requirement #2: For convexity, a shift size of 25 bp is recommended to ensure reasonable convergence for 256 paths.



3. Requirement #3: For partial volatility durations, a shift size of 50 bp for the *Black vol (points)* and *bp vols (annual)*, and a shift size of 0.5% for *Black vol (%)* are recommended to ensure reasonable convergence for 256 paths.

Model Input

1. Main External Static Data Source: The main external source for static data – bond details – is readily available from Bloomberg. For newly created securities without static data, appropriate proxy will be applied.
2. Securities with Missing Inputs: Standard assumptions of WALA, WAC, and WAM will be applied if a newly created security is missing such information during the first month after its creation.
3. Primary Mortgage Rate: The Freddie Mac weekly survey of mortgage rates is the industry standard for the primary mortgage rate PMMS – Primary Mortgage Market Survey.
4. Primary Mortgage Rate 30Y Index: Since 1972, Freddie Mac has been reporting average 30-year rates and points paid compiled from a number of commercial banks, thrift institutions, and mortgage lenders across the country.
5. Primary Mortgage Rate 15Y Index: A similar series for 15Y rates spans 1991 to present. This is obtained from Bloomberg.
6. Current Coupon Rate: The industry standard is to take the TBA benchmark contract – its market observable closing price – for the next two months and linearly interpolate to solve for the par dollar price equivalent coupon rate for 30-day forward using the day count.

Model Output

1. Key Rate Duration: 3M, 6M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 15Y, 20Y, 30Y. Price change resulting from a 100 bp shift to the specified key rate.



$$KRDur = \frac{PV_{down} - PV_{up}}{2 \times PV} \times \frac{10,000}{shift}$$

$$KR \frac{DP}{DY} = KRDur \times \frac{PV}{100}$$

2. Key Rate Convexity: 3M, 6M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 15Y, 20Y, 30Y. Partial convexity, expressed as percentage market value change per 100 bp².

$$KRCvx = \frac{(PV_{down} - PV) - (PV_{up} - PV)}{PV \times \left(\frac{shift}{10,000}\right)^2 \times 100}$$

$$KR Dlr Cvx = \frac{KRCvx \times PV \times Holding}{10,000}$$

3. Key Rate DV01: 3M, 6M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 15Y, 20Y, 30Y. Partial DV01, expressed as percentage market value change per 100 bp.

$$KR DV01 = \frac{KRCur \times PV \times Holding}{1,000,000}$$

$$KR Dlr Dur = KR DV01 \times 100$$

4. Volatility Dollar Duration: Market value change given a percentage point shift in volatility.
 5. Volatility Duration: Percentage change in market value given a percentage point shift in volatility.
 6. Vega: Dollar change in price of an option given a percentage point shift in volatility.



Mortgage Spread Duration

1. Mortgage Spread Duration: Market value change given a 100 bp change in mortgage spread.

$$Mtge Sprd Dur = \frac{PV_{down} - PV_{up}}{2 \times PV} \times \frac{10,000}{shift}$$

2. Mortgage Spread DV01: Dollar value change in price given a 1 bp change in mortgage spread.

$$Mtge Sprd DV01 = \frac{Mtge Sprd \times PV}{1,000,000}$$

3. Mortgage Spread Dollar Duration: Market value change given a 100 bp change in mortgage spread.

$$Mtge Sprd Dlr Dur = Mtge Sprd DV01 \times Holding$$

4. Mortgage Spread Convexity:

$$Mtge Sprd Cvx = \frac{(PV_{down} - PV) - (PV_{up} - PV)}{PV \times \left(\frac{shift}{10,000}\right)^2} \times 100$$

5. Mortgage Spread Convexity 01:

$$Cvx Mtge Sprd 01 = \frac{Mtge Sprd Cvx \times PV}{10,000}$$



6. Mortgage Spread Dollar Convexity: Market value change given a 100 bp change in mortgage spread.

$$Mtge\ Sprd\ Dlr\ Cvx = Mtge\ Sprd\ Cvx\ 01 \times Holding \times 100$$

Option-Adjusted Spread Duration

1. The OAS Measure: The option-adjusted spread – OAS – is a measure used to value securities with embedded options. It is calculated for mortgage-related securities using Monte-Carlo simulation.
2. Cashflows over the Monte-Carlo Path: In this approach, randomly generated interest-rate paths are projected over the life of a mortgage security to simulate the impact of changing interest rates on prepayment and on the PV's of the resulting cashflows.
3. PV for each Path: A PV is obtained for each path by discounting the mortgage cashflows using the forward rates along the path – the reference curve can be UST, LIBOR, or Agency.
4. Mortgage Yields vs UST: Since mortgage yields are higher than UST/LIBOR, the paths PV's will, on average, exceed the mortgage price.
5. Calculating OAS over Paths: Using iteration, one solves for the required spread over each cashflow's discount rate in order to equate the average path PV with the PV of the mortgage. That spread is the OAS.
6. ZVol OAS: Zero volatility OAS.
7. Option Cost: Difference between OAS and Zero Volatility OAS - $ZVolOAS - OAS$
8. Option Adjusted Average Life: The average of the Path Average Life's generated within the OAS framework.
9. ZVol Average Life: Zero Volatility Average Life
10. ZVol Yield: Zero volatility yield
11. Option Adjusted Spread Duration: Percentage change in market value given a 100 bp change in OAS per par.



$$OA\ Sprd\ Dur = \frac{PV_{down} - PV_{up}}{2 \times PV} \times \frac{10,000}{shift}$$

12. OAS DV01: Dollar value change in market value for a 1 bp change in OAS per par.

$$DV\ OAS01 = \frac{OA\ Sprd\ Dur \times PV}{10,000}$$

13. Total OAS01: Dollar value change in market value for a 1 bp change in OAS.

$$Total\ OAS01 = \frac{DVOAS01 \times Holding}{100}$$

14. Spread Dollar Duration: Market value change given a 100 bp change in OAS.

$$Sprd\ Dlr\ Dur = Total\ OAS01 \times 100$$

15. OAS Convexity: Percentage change in market value due to convexity given a 100 bp² change in OAS

$$OA\ Sprd\ Cvx = \frac{(PV_{down} - PV) - (PV_{up} - PV)}{PV \times \left(\frac{shift}{10,000}\right)^2} \times 100$$

16. OA Spread Convexity 01: Dollar value change in price due to convexity given a 1 bp² change in OAS per par

$$Cvx\ OAS01 = \frac{OA\ Sprd\ Cvx \times PV}{1,000,000}$$



17. Total OAS Convexity 01: Dollar value change in market value convexity for a 1 bp² change in OAS.

$$Total\ Cvx\ OAS01 = \frac{Cvx\ OAS01 \times Holding}{100}$$

18. Spread Dollar Convexity: Market value change given a 100 bp² change in mortgage spread.

$$Sprd\ Dlr\ Cvx = Total\ Cvx\ OAS01 \times 100^2$$

Prepay Duration

1. Definition: Prepay duration measures the sensitivity to prepayment projections. It is calculated by generating the market values corresponding to prepay multipliers of 1.05 and 0.95, finding the difference between these two values, and expressing the difference as a market value sensitivity to a 10% change in prepayment speeds.
2. Prepay Duration Calculation: Percentage change in market value given a 10% change in prepayment speed under a constant OAS assumption.

$$Prepay\ Dur = 100 \times \frac{0.1}{shift} \times \frac{PV_{down} - PV_{Up}}{2 \times PV}$$

3. Prepay Dollar Duration Calculation: Market value change given a 10% change in prepayment speed under a constant OAS assumption.

$$Prepay\ Dlr\ Dur = \frac{Prepay\ Dur \times Market\ Value}{100}$$



4. Prepay Convexity Calculation: Captures the asymmetry of the market value change.

$$Prepay\ Cvx = \frac{(PV_{down} - PV) - (PV_{up} - PV)}{PV \times \left(\frac{shift}{0.1}\right)^2}$$

where *shift* equals 0.05 – 5% shift up and 5% shift down.

5. Total Prepay DV01: Dollar value change in market value for 1% change in prepayment speed.

Model Design and Assumptions

1. Forward Rate Volatility Function Specification: PolyPaths specifies the forward rate volatility

$$\sigma_{FWD}(t, T) = A(T - t)B(T)C(t)$$

which is a typical specification where $A(x)$ is the homogenous volatility that only depends on time-to-maturity, the leverage $B(x)$ and the time volatility $C(x)$ are elastic corrections used to smooth out the volatility structure.

2. $A(x)$ when $x \leq 15$: When

$$x \leq 15$$

$A(x)$ is a piece-wise linear function with 9 parameters positioned at

$$x = 0, 0.5, 1, 2, 3, 5, 7, 10, 15$$



3. $A(x)$ when $x > 15$: When

$$x > 15$$

$$A(x) = A(15)e^{-const(x-15)}$$

4. Calibration Instruments in BGM: The BGM 3 Normal Model – as well as the Displaced Diffusion Model – are calibrated only with swaptions and not caplets.
5. Standard Calibration Set: The standard calibration set used for models is (1, 2, 3, 5, 7, 10) by (2, 3, 5, 10) swaptions.
6. Calibration Instruments/Scheme for SOFR: For SOFR, the default setting does not include caplets in the calibration. During calibration, the system attempts to minimize the sum of the squared errors for the differences between the model implied volatilities and their market counterparts.

Pricing – Mathematical Background

1. Framework for CMO Pricing: The following section provides the mathematical background for different types of CMOs. The description provides theoretical framework for pricing and cashflow generation.
2. Customization Employed by Intex/PolyPaths: The model described below is generic in nature. The underlying model used within PolyPaths/Intex will broadly conform to the model described below but is likely to differ in various aspects – calibration, implementation, etc.
3. $\underline{x}$: Risk Factor which affects the MBS prices and follows a continuous random process having:
4. $\underline{\mu}$: Drift rate
5. $\underline{\sigma}$: Volatility



6. $\underline{z}(t)$: Standard Brownian motion
7. $\underline{c}$: Coupon rate
8. $\underline{B}(t)$: Balance at time t
9. $\underline{\lambda}(t)$: Amortization rate
10. $\underline{V}$: Total Value of the MBS
11. $\underline{R}$: Expected Annualized Return
12. $\underline{r}$: Risk-free rate
13. Evolution of x :

$$\Delta x = \mu \Delta t + \sigma \Delta z$$

14. MBS Price P : The MBS price P can be viewed as a function of time and risk factors, i.e.

$$P \equiv P(t, x)$$

Applying Ito's lemma

$$\Delta P = \left(\frac{\partial P}{\partial t} + \mu \frac{\partial P}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 P}{\partial x^2} \right) \Delta t + \sigma \frac{\partial P}{\partial x} \Delta z$$

$$V = PB$$

$$\mathbb{E}[\Delta V] = \mathbb{E}[\Delta(PB)] = \mathbb{E}[P\Delta B + B\Delta P] = B \left(\frac{\partial P}{\partial t} + \mu \frac{\partial P}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 P}{\partial x^2} \right) \Delta t - \lambda PB \Delta t$$

15. Interest and Principal Flows: The investment will accrue interest cash $cB\Delta t$ and principal cash $\lambda B\Delta t$ and it is assumed that the latter is paid back.
16. The Annualized Return: Adding the two terms to the equation and dividing by $PB\Delta t$, one obtains the expected annualized return.



17. Composite Portfolio Return: This gives

$$RP = \frac{\partial P}{\partial t} + \mu \frac{\partial P}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 P}{\partial x^2} + c + \lambda - \lambda P$$

18. Composite IO Portfolio Return: For IO, the above equation can be modified to remove the principal term

$$RP = \frac{\partial P}{\partial t} + \mu \frac{\partial P}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 P}{\partial x^2} + c - \lambda P$$

19. Decomposing R into r and OAS : The expected return R can be considered to be the sum of the risk-free rate r and an option-adjusted spread OAS . OAS can be considered as an additional return compensating for the risks not fully reflected in the dynamics of the modeled risk factors.

20. SDE for the OAS : If the MBS market price $P(0, x(0))$ is known today, one can solve for the OAS as

$$(r + OAS)P = \frac{\partial P}{\partial t} + \mu \frac{\partial P}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 P}{\partial x^2} + c - \lambda P$$

21. Interest Rates Stochastic Equations: The interest rate stochastic equations of either the Displaced Diffusion or the Normal Model are those of the Normal Model are similar to those of the lognormal of the CEV model (Hull and White (2000)) except for the different volatility terms.

22. Displaced Diffusion LIBOR Market Model: For the Displaced Diffusion LIBOR Market Model, also known as the Shifted Lognormal Model (Hull and White (2000)), one wants to capture the correct dynamics of the interest rates, and the skew observed for the OTM swaptions and the caplets.



23. Corresponding Forward Rate Dynamics: The forward rate dynamics, under the measure $\mathcal{M}[P(t, t_m)]$ with $P(t, T)$ being the zero-coupon bond price and $m(t)$ being the index for the next reset date at time t , are represented by:

$$\Delta F_{\kappa}(t) = [F_{\kappa}(t) + \alpha] \sum_{i=m(t)}^{\kappa} \frac{\delta_i [F_i(t) + \alpha] \sum_{q=1}^{p=3} \sigma_{iq}(t) \sigma_{\kappa q}(t)}{1 + F_i(t) \sigma_i} \Delta t + \sum_{q=1}^3 \sigma_{\kappa q}(t) [F_{\kappa}(t) + \alpha] \Delta z_q$$

where $F_{\kappa}(t)$ is the forward rate observed at time t for the period $(t_{\kappa}, t_{\kappa+1})$, expressed with a compounding period of δ_{κ} , α the constant shift, Δz_q is the q^{th} independent Brownian, $\sigma_{\kappa q}(t)$ is the q^{th} component of the Displaced Diffusion volatility, noting that the model is a 3-factor model.

24. Intuition behind α : With

$$\alpha = 0$$

the above equation is the lognormal model, while with α being a large value, such as 100%, it behaves like a normal model.

25. Crossing the Zero Forward-Rate Barrier: Note that this model is also referred to as the shifted lognormal model: if the initial rate is above $-\alpha$, then the projected rate is always above $-\alpha$, just like interest rates are always above 0 in the lognormal model.

26. Normal LMM Forward Rate Dynamics: For the Normal LMM, the forward rate dynamics are represented by

$$\Delta F_{\kappa}(t) = \sum_{i=m(t)}^{\kappa} \frac{\delta_i \sum_{q=1}^{p=3} \sigma_{iq}(t) \sigma_{\kappa q}(t)}{1 + F_i(t) \delta_i} \Delta t + \sum_{q=1}^3 \sigma_{\kappa q}(t) \Delta z_q$$

where $F_{\kappa}(t)$ is the forward rate, $\sigma_{\kappa q}(t)$ the q^{th} component normal volatility.



$$p = 3$$

for this 3-factor model.

27. Displaced Diffusion Option Pricer Setup: Following Hull and White (2000), consider a European swaption that lasts from t_n to t_{N+1} with a swap rate $S_{n,N}(t)$. The strike is R_s and $\sigma_{n,N}$ is the standard deviation in the corresponding model.

28. Swap Annuity: It is defined that

$$A_{n,N}(t) = \sum_{i=n}^N \delta_i P(t, t_{i+1})$$

29. The corresponding Swap Rate: Then the swap rate is

$$S_{n,N}(t) = \frac{P(t, t_n) - P(t, t_{N+1})}{A_{n,N}(t)}$$

30. Displaced Diffusion Swap Rate Dynamics: Using $A_{n,N}(t)$ as the numeraire, the swap rate dynamics for the Displaced Diffusion model is

$$\Delta S_{n,N}(t) = [S_{n,N}(t) + \alpha] \sigma_{n,N} \Delta z$$

where $\sigma_{n,N}$ is the Displaced Diffusion volatility of the swap rate.

31. Displaced Diffusion European Swaption Price: The swaption price for the Displaced Diffusion model is

$$f(0) = L_s A_{n,N}(0) \{ [S_{n,N}(0) + \alpha] \mathcal{N}(d_1) - (R_s + \alpha) \mathcal{N}(d_2) \}$$



where L_s is the swap principal, $\sigma_{n,N}$ is the volatility of the swaption in the Displaced Diffusion Model, and

$$d_1 = \frac{\ln \frac{S_{n,N}(0) + \alpha}{R_s + \alpha} + \frac{\sigma_{n,N}^2 t_n}{2}}{\sigma_{n,N} \sqrt{t_n}}$$

$$d_2 = d_1 - \sigma_{n,N} \sqrt{t_n}$$

32. Normal Mode Swap Rate Dynamics: Similarly for the Normal model, the swap rate $S_{n,N}(t)$ follows

$$\Delta S_{n,N}(t) = \sigma_{n,N} \Delta z$$

where $\sigma_{n,N}$ is the normal volatility of the swap rate.

33. Normal Model European Swaption Price: The swaption price for the Normal model is given by

$$f(0) = L_s A_{n,N}(0) \left\{ [S_{n,N}(0) - R_s] \mathcal{N}(d_1) - \frac{\sigma_{n,N} \sqrt{t_n}}{\sqrt{2\pi}} e^{-\frac{d_1^2}{2}} \right\}$$

where $\sigma_{n,N}$ is the swaption volatility in the Normal model, and

$$d_1 = \frac{S_{n,N}(0) - R_s}{\sigma_{n,N} \sqrt{t_n}}$$

34. Black Volatility Standardization across Models: The above swaption formula will be used to convert among Black volatility, Displaced Diffusion volatility, and Normal volatility in the calibrations of LMM.



35. Advantage of bp Volatility Marking: Both the Displaced Diffusion and the Normal LIBOR market models can simulate positive and negative interest rate scenarios, but the definition of Black implied volatilities is no longer available for negative swap rates, while bp volatilities are well-defined and sensible. Hence, the two models will target bp vols in calibrations.

36. Monte Carlo of Forward Rates: Following

$$\Delta F_{\kappa}(t) = [F_{\kappa}(t) + \alpha] \sum_{i=m(t)}^{\kappa} \frac{\delta_i [F_i(t) + \alpha] \sum_{q=1}^{p=3} \sigma_{iq}(t) \sigma_{\kappa q}(t)}{1 + F_i(t) \sigma_i} \Delta t + \sum_{q=1}^3 \sigma_{\kappa q}(t) [F_{\kappa}(t) + \alpha] \Delta z_q$$

one can generate the forward-rate paths via a Monte-Carlo simulation:

$$F_i(t_{\kappa+1}) + \alpha = [F_i(t_{\kappa}) + \alpha] e^{\left\{ \sum_{i=m(t)}^{\kappa} \frac{\delta_i [F_i(t) + \alpha] \sum_{q=1}^{p=3} \sigma_{iq}(t_{\kappa}) \sigma_{\kappa q}(t_{\kappa})}{1 + F_i(t) \sigma_i} - \frac{1}{2} \sum_{q=1}^{p=3} \sigma_{iq}^2(t_{\kappa}) \right\} \delta_{\kappa} + \left[\sum_{q=1}^{p=3} \sigma_{iq}(t_{\kappa}) \epsilon_i \right] \sqrt{\delta_{\kappa}}}$$

where ϵ_i is the random sample from the standard normal distribution.

37. Forward-rate Displaced Diffusion Volatility: The forward rate displaced diffusion volatility $\sigma_i(t)$ determines the extent of the fluctuations of the generated rates, and should be calibrated to the market observables.

38. Market Observables of the Volatilities: Note that the market observables are the swaption Black vols or bp vols, and one needs to convert them into swaption Displaced Diffusion vols using

$$\Delta S_{n,N}(t) = [S_{n,N}(t) + \alpha] \sigma_{n,N} \Delta z$$

39. Simultaneous Calibration Displaced Diffusion Volatilities: In the current model, one calibrates $\sigma_i(t)$ to the swaption Displaced Diffusion vols simultaneously. One first needs to derive the formulas of Swaption Displaced Diffusion vols under the LMM framework.



40. Displaced Diffusion Swaption Vol Expression: For the expression of the swaption vol, an approximation is used:

$$V_{N,M}^2(t) = \frac{1}{T_N} \sum_{i,j=N+1}^M \frac{\rho_{ij} w_i(0) w_j(0) [F_i(0) + \alpha] [F_j(0) + \alpha]}{[S_{N,M}(0) + \alpha]^2} \int_0^{T_N} \sigma_i(t) \sigma_j(t) dt$$

where $S_{N,M}(t)$ is the forward swap rate, and can be thought of as an algebraic transformation of forward rates:

$$S_{N,M}(t) = \sum_{i=N+1}^M w_i(t) F_i(t)$$

and

$$w_i(t) = \frac{\delta_i \prod_{j=N+1}^i \frac{1}{1 + \delta_j F_j(t)}}{\sum_{\kappa=N+1}^M \delta_{\kappa} \prod_{j=N+1}^{\kappa} \frac{1}{1 + \delta_j F_j(t)}}$$

41. Adjustment of the Forward Vols: Having the above expressions in mind, one adjusts the forward rate Displaced Diffusion volatility $\sigma_i(t)$ so that the errors between the Model vols and the corresponding market Displaced Diffusion vols are minimized.

42. Expression for Normal Swaption Volatility: For the Normal vol, with $\sigma_i(t)$ the bp vol, one simply modifies:

$$V_{N,M}^2(t) = \frac{1}{T_N} \sum_{i,j=N+1}^M \frac{\rho_{ij} w_i(0) w_j(0) [F_i(0) + \alpha] [F_j(0) + \alpha]}{[S_{N,M}(0) + \alpha]^2} \int_0^{T_N} \sigma_i(t) \sigma_j(t) dt$$

into



$$V_{N,M}^2(t) = \frac{1}{T_N} \sum_{i,j=N+1}^M \rho_{ij} w_i(0) w_j(0) \int_0^{T_N} \sigma_i(t) \sigma_j(t) dt$$

43. Truncated Newton Algorithm: PolyPaths uses the truncated Newton algorithm (Nash (1984)) as the optimizer used in the calibration.

References

- Brace, A., D. Gatarek, and M. Musiela (1997): The Market Model of Interest Rate Dynamics *Mathematical Finance* **7 (2)** 127-155
- Hayre, L. (2001): *Solomon Smith Barney Guide to Mortgage-Backed and Asst-Backed Securities* **John Wiley and Sons** Hoboken NJ
- Hull, J. C., and A. D. White (2000): Forward Rate Volatilities, Swap Rate Volatilities, and Implementation of the LIBOR Market Model *Journal of Fixed Income* **10 (2)** 46-62
- Nash. S (1984): Newton-type Minimization via the Lanczos Method *SIAM Journal on Numerical Analysis* **21 (4)** 770-788